Monogamy relations for symmetric multiqubit systems

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Abstract

New results

- Some relations among the quantities $\mathcal{A}_t$, r_t, S_t, etc.
- Inequalities (I2). In particular, we have found a way to prove the r.h.s of the equation i) algebraically for up to $j = 3$, and ii) numerically for up to $j = 100$.
- Sharp global bounds on S_8 for pure symmetric nine-qubit states, including an algebraic maximizing state and exact sum-of-squares certificates for both extrema; see Theorem 1.

Open questions

- Which inequalities can we recover from the sym to the nonsymmetric states?
- What are the differences of $\mathcal{S}$ for the set of general state and symmetric states?
- Write all the inequalities as in R in our paper, but now for symmetric and adding the new inequalities.
- For the symmetric case, write the S_m in terms of the Majorana stars.
- Check if the l.h.s. of Eq. (I3) is easy to prove.

I. INTRODUCTION

A. Non-symmetric case

We begin by considering a state ρ of a system of N qubits with Hilbert space of quantum states denoted by $\mathcal{H}^{\otimes N}$, with $\mathcal{H}$ the Hilbert space of a single qubit. Its expansion in the Pauli string basis is

$$\rho = \frac{1}{2^N} \sum_{\boldsymbol{\mu}} x_{\boldsymbol{\mu}} \sigma_{\boldsymbol{\mu}} \,, \tag{1}$$

with $x_{\boldsymbol{\mu}} = \mathrm{Tr}(\rho\sigma_{\boldsymbol{\mu}})$, $\boldsymbol{\mu} = \mu_1\mu_2\ldots\mu_N$ a multi-index and

$$\sigma_{\boldsymbol{\mu}} = \sigma_{\mu_1\mu_2\ldots\mu_N} \equiv \sigma_{\mu_1} \otimes \cdots \otimes \sigma_{\mu_N} \,, \tag{2}$$

where σ_0 denotes the identity operator and σ_a ($a = 1, 2, 3$) are the Pauli matrices. Each index μ_α takes values in $\{0, 1, 2, 3\}$. This parametrization has the advantage that the reduced density matrices of the original state are parametrized by subsets of the $x_{\boldsymbol{\mu}}$ parameters. Indeed, consider a subset $A \subset \{1, \ldots, N\}$ of k qubits, and the corresponding k-qubit reduced state ρ_A obtained by tracing out the complement of A, $\bar{A}$. The expansion (1) of ρ_A has coefficients $x^A_{\mu_1\mu_2\ldots\mu_k}$ given by those of ρ with zeros placed on the indices associated with $\bar{A}$ [1, 2]. For instance, if $A = \{1, \ldots, k\}$, then

$$x^A_{\mu_1\ldots\mu_k} = x_{\mu_1\ldots\mu_k 0\ldots 0} = x_{\mu_1\ldots\mu_k \mathbf{0}_{N-k}} \tag{3}$$

where $\mathbf{0}_{N-k}$ denotes a string of $N - k$ zeros.

Let us now define quantities related to m-body correlations. Given a Pauli string, $\sigma_{\boldsymbol{\mu}}$, the number of non-zero indices in $\boldsymbol{\mu}$ [3] is called its *weight* and denoted by $\mathrm{wt}(\sigma_{\boldsymbol{\mu}})$. The sector length $S_m(\rho)$ is defined as the sum [4–6],

$$S_m(\rho) \equiv \sum_{\mathrm{wt}(\sigma_{\boldsymbol{\mu}})=m} x^2_{\boldsymbol{\mu}}. \tag{4}$$

Another quantity of interest would be the purity of a reduced qubit state ρ_A. If $|A| = t$, then

$$\mathrm{Tr}(\rho_A^2) = \frac{1}{2^t} \sum_{\boldsymbol{\mu}=\mu_1\ldots\mu_t} (\mathrm{Tr}\rho_A\sigma_{\boldsymbol{\mu}})^2 \,, \tag{5}$$

We can now calculate the average purity r_t with respect to all the subsystems of order t

$$r_t(\rho) \equiv \binom{N}{t}^{-1} \sum_{|A|=t} \mathrm{Tr}(\rho_A^2), \tag{6}$$

which is a quantity invariant under local unitary transformations.

B. Symmetric sector

A (not-necessarily pure) state ρ of an N-qubit system is *fully* symmetric if $\pi_{ij}\rho = \rho\pi_{ij} = \rho$ for any exchange-

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particle operation π_{ij} that swaps the i-th and j-th qubits. Due to the symmetrization of the state, $x_{\boldsymbol{\mu}} = x_{\boldsymbol{\nu}}$ for any pair of multi-indices $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ having the same multiplicities of the four types of values, $0, 1, 2, 3$. Additionally, $\rho_A = \rho_{A'}$ for any pair of subsets A and A' of the same size. The fully symmetric states are spanned by the symmetric Dicke states

$$|D_N^{(\alpha)}\rangle = \binom{N}{\alpha}^{-1/2} \sum_{\sigma} \pi_\sigma |\mathbf{0}_{N-\alpha}\mathbf{1}_\alpha\rangle \tag{7}$$

for $\alpha = 0, \ldots, N$, where the sum runs over the $\binom{N}{\alpha}$ distinct permutations of the bit string, each counted once. These states span the symmetric sector of N qubits denoted by $\mathcal{H}^{\vee N} \subset \mathcal{H}^{\otimes N}$. Throughout the text, we specify symmetric states or quantities restricted to the symmetric sector with the label sym. The projector onto $\mathcal{H}^{\vee N}$ is given by

$$\Pi_{\text{sym}} \equiv \sum_{\alpha=0}^{N} |D_N^{(\alpha)}\rangle \langle D_N^{(\alpha)}| . \tag{8}$$

A more suitable representation for symmetric state ρ is the tensor representation introduced in [1]. In this representation, a symmetric state is written as

$$\rho = \frac{1}{2^N} \sum_{\boldsymbol{\mu}} x_{\boldsymbol{\mu}} L_{\boldsymbol{\mu}}, \qquad x_{\boldsymbol{\mu}} = \text{Tr}\left(\rho L_{\boldsymbol{\mu}}\right). \tag{9}$$

In Eq. (9), the $L_{\boldsymbol{\mu}}$ are the symmetrized Pauli strings $L_{\boldsymbol{\mu}} \equiv \Pi_{\text{sym}} \sigma_{\boldsymbol{\mu}} \Pi_{\text{sym}}$, and can be obtained by expanding the square of the $(N+1)$-dimensional matrix corresponding to the $(N/2, 0)$ representation of a Lorentz boost [1]. The 4^N operators $L_{\boldsymbol{\mu}}$ are invariant under permutation of their indices; note that the $x_{\boldsymbol{\mu}}$ in (9) are the same as in (1) since ρ is symmetric.

Another suitable basis is given by the multipole operators $T_{LM}^{(N)}$ [7, 8] defined here in terms of the Dicke states (see Appendix C). The expansion of ρ in this basis reads

$$\rho = \sum_{L=0}^{N} \sum_{M=-L}^{L} \rho_{LM} T_{LM}^{(N)}, \quad \text{with } \rho_{LM} = \text{Tr}(\rho T_{LM}^{(N)\dagger}) . \tag{10}$$

for $L = 0, \ldots, N$ and $M = -L, \ldots, L$. A possible expansion of the $T_{LM}^{(N)}$ in terms of the $\tilde{\sigma}_{\boldsymbol{\mu}}$ is given in Ref. C. Whenever there is no possible confusion, we omit the superscript (N).

For fully symmetric states, we have that

$$\begin{aligned} S_m &= \sum_{\text{wt}(\boldsymbol{\mu})=m} x_{\boldsymbol{\mu}}^2 \\ &= \binom{N}{m} \sum_{\alpha\,\beta\,\gamma} \binom{m}{\alpha\,\beta\,\gamma} x^2_{(\mathbf{0}_{N-m}\mathbf{1}_\alpha\mathbf{2}_\beta\mathbf{3}_\gamma)}, \end{aligned} \tag{11}$$

where

$$\binom{m}{\alpha\,\beta\,\gamma} \equiv \frac{m!}{\alpha!\beta!\gamma!} \tag{12}$$

are the multinomial coefficients. The multiplicities are nonnegative integers; the coefficient is zero unless $\alpha + \beta + \gamma = m$. Throughout, $\binom{n}{k} = 0$ for integers $n \geqslant 0$ and $k \notin \{0, \ldots, n\}$. Similarly, the r_t coefficients reduce in the symmetric sector to elements of a single t-qubit reduced state ρ_A

$$\begin{aligned} r_t\left(\rho_{\text{sym}}\right) &= \frac{1}{2^t} \sum_{\boldsymbol{\mu}} \left(\text{Tr}\rho_A \sigma_{\boldsymbol{\mu}}\right)^2 \\ &= \frac{1}{2^t} \sum_{\alpha\,\beta\,\gamma\,\delta} \binom{t}{\alpha\,\beta\,\gamma\,\delta} \left(\text{Tr}\rho_A \sigma_{(\mathbf{0}_\delta\mathbf{1}_\alpha\mathbf{2}_\beta\mathbf{3}_\gamma)}\right)^2 \\ &= \frac{1}{2^t} \sum_{\alpha\,\beta\,\gamma\,\delta} \binom{t}{\alpha\,\beta\,\gamma\,\delta} \left(\text{Tr}\rho \sigma_{(\mathbf{0}_{N-t+\delta}\mathbf{1}_\alpha\mathbf{2}_\beta\mathbf{3}_\gamma)}\right)^2 . \end{aligned} \tag{13}$$

This quantity has been used in Ref. [9]. By definition, it is constrained to the values

$$\frac{1}{m+1} \leqslant r_m(\rho_{\text{sym}}) \leqslant 1. \tag{14}$$

In addition to these quantities, there are others defined originally only for symmetric states. For instance,

$$a_L(\rho) = \sum_{M=-L}^{L} |\rho_{LM}|^2. \tag{15}$$

In particular, $a_0(\rho) = (N+1)^{-1}$ and the purity of ρ is

$$\text{Tr}\left[\rho^2\right] = \sum_{L=0}^{N} a_L. \tag{16}$$

In Appendix D, we present a table with all the quantities mentioned here and others relevant in the symmetric case.

As in the non-symmetric case, we define the time-reversed states

$$\tilde{\rho} \equiv \sigma_2^{\otimes N} \rho^* \sigma_2^{\otimes N}, \tag{17}$$

and equivalently for pure states

$$|\tilde{\psi}\rangle = \sigma_2^{\otimes N} |\psi^*\rangle . \tag{18}$$

II. EXPLICIT RELATIONS BETWEEN THE INVARIANT QUANTITIES

We now present relations among the variables r_t, S_m, and a_L. Some derivations are given in Appendix E. First, we start with the relations between r_t and S_m (Appendix E 1).

$$r_t = \frac{1}{2^t} \sum_{m=0}^{t} \frac{\binom{t}{m}}{\binom{N}{m}} S_m , \tag{19}$$

or, by binomial inversion,

$$S_m = \binom{N}{m} \sum_{t=0}^{m} (-1)^{t+m} 2^t \binom{m}{t} r_t. \tag{20}$$

For the symmetric states, r_t in terms of a_L reads [10, 11]

$$r_t = \sum_{L=0}^{t} \frac{\binom{t}{L}\binom{N+L+1}{N}}{\binom{N}{L}\binom{t+L+1}{t}} a_L, \tag{21}$$

which can be inverted (see Appendix E 2) as

$$a_L = \frac{(2L+1)\binom{N}{L}}{(L+1)\binom{N+L+1}{N}} \sum_{t=0}^{L} (-1)^{t+L} \binom{L}{t}\binom{t+L}{t} r_t. \tag{22}$$

This entails the relationships between S_m and a_L (see Appendix E 3)

$$S_m = \binom{N}{m} \sum_{\substack{0\leqslant L\leqslant m \\ m+L \text{ even}}} 2^L \frac{\binom{N+L+1}{N}}{\binom{N}{L}\binom{m+L+1}{m}} \binom{\frac{m+L}{2}}{\frac{m-L}{2}} a_L \tag{23}$$

and

$$\begin{aligned} a_L =& \frac{(2L+1)\binom{N}{L}}{2^L(L+1)\binom{N+L+1}{N}} \\ &\times \sum_{\substack{0\leqslant m\leqslant L \\ m+L \text{ even}}} \frac{(-1)^{\frac{m-L}{2}}}{\binom{N}{m}} \binom{L+m}{\frac{L+m}{2}} \binom{\frac{L+m}{2}}{\frac{L-m}{2}} S_m. \end{aligned} \tag{24}$$

III. OLD INEQUALITIES FOR PURE SYMMETRIC STATES

For pure states, we enlist some inequalities proved in several references for the S_m coefficients

$$\begin{aligned} &\sum_{m=0}^{N} S_m = 2^N \\ &0 \leqslant \sum_{m=0}^{N} (-1)^m S_m \leqslant \left(1+(-1)^N\right) 2^{N-1} \\ &\frac{1}{2^{N-k}} \sum_{m=0}^{N-k} \binom{N-m}{k} S_m = \frac{1}{2^k} \sum_{m=0}^{k} \binom{N-m}{N-k} S_m \end{aligned} \tag{25}$$

The coefficients S_m can be seen as expectation values of double copy quantities, giving rise to the well-known shadow inequalities [12]

$$\sum_{m=0}^{N} (-1)^m K_k(m,N;4) S_m \geqslant 0, \tag{26}$$

for all $k = 0,\ldots,N$ and where $K_k(n,N;4)$ are the Kravchuk polynomials (C4).

Other inequalities beyond the the shadow inequalities were derived in Ref. [12] by using properties of the Schmidt decompositions of pure states. They are given by

$$\begin{aligned} &\frac{2^k}{2^{\min(k,N-k)}} \binom{N}{k} \leqslant \sum_{m=0}^{k} \binom{N-m}{k-m} S_m \leqslant 2^k \binom{N}{k}, \\ &0 \leqslant \sum_{m=0}^{k} (-1)^m \binom{N-m}{k-m} S_m \leqslant 2^k \binom{N}{k}, \\ &\sum_{m=0}^{\lfloor \frac{k-1}{2} \rfloor} \binom{N-2m}{k-2m} S_{2m} \leqslant 2^{k-2} \left(3+(-1)^k\right) \binom{N}{k}. \end{aligned} \tag{27}$$

The objective of this work is the comparison and calculation of the numerical range of the sector lengths of pure generic states and pure symmetric states. By considering the sector length vector $\mathbf{S} \equiv (S_1,\ldots,S_N) \in \mathbb{R}_+^N$, we define the *numerical range* of sector lengths for pure states for general and symmetric states

$$\begin{aligned} \mathcal{S} &\equiv \left\{ \mathbf{S}(\rho) \,\middle|\, \rho = |\psi\rangle\langle\psi| \right\}. \\ \mathcal{S}^{\text{sym}} &\equiv \left\{ \mathbf{S}(\rho) \,\middle|\, \rho = |\psi\rangle\langle\psi| \text{ and } |\psi\rangle \in \mathcal{H}^{\vee N} \right\}. \end{aligned} \tag{28}$$

respectively.

For non-necessarily symmetric states, we define the subset of $\mathbb{R}_+^N$ specified by the previous inequalities mentioned above

$$\mathcal{R} = \left\{ \mathbf{S} \,\middle|\, \text{Eqs. (25),(26) and (27) hold } \forall\, k = 0,\ldots,N \right\}. \tag{29}$$

The sets of inequalities (25) and (26) can give inequivalent constraints, as it is discussed in Ref. [12].

A. (Optional) Extra equalities written in the other variables

$$\sum_{L=0}^{N} a_L = 1, \tag{30}$$

In terms of the other variables, the last equality is equal to

$$r_k = r_{N-k}, \tag{31}$$

IV. NEW FAMILIES OF INEQUALITIES FOR PURE SYMMETRIC STATES

A. Inequalities from the double-copy technique

Here, we derive new inequalities for the sector lengths of symmetric qubits states. We employ a similar double-copy technique as used to derive the shadow inequalities in Ref. [12]. Our quantity of merit are the variables a_L's,

which for a pure symmetric state $|\psi\rangle \in \mathcal{H}^{\vee N}$ can be written as an expectation value in the double-copy Hilbert space

$$a_L = \langle\Psi| \mathcal{T}_L |\Psi\rangle , \tag{32}$$

with $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle \in \left(\mathcal{H}^{\vee N}\right)^{\otimes 2}$ and

$$\mathcal{T}_L \equiv \sum_{M=-L}^{L} T_{LM} \otimes T_{LM}^{\dagger}. \tag{33}$$

The eigendecomposition of these operators are derived by using techniques of coupling angular momentum in Ref. [13] (see also Appendix F),

$$\mathcal{T}_L = \sum_{j=0}^{N} \lambda_j^{(L)} \mathcal{P}_j \tag{34}$$

with

$$\begin{aligned} \lambda_j^{(L)} \equiv & (2L+1)(-1)^{N+j} \begin{Bmatrix} N/2 & N/2 & j \\ N/2 & N/2 & L \end{Bmatrix} \\ \mathcal{P}_j \equiv & \sum_{m=-j}^{j} |\phi_{jm}^{\text{2-copy}}\rangle \langle\phi_{jm}^{\text{2-copy}}| , \end{aligned} \tag{35}$$

and[1]

$$|\phi_{jm}^{\text{2-copy}}\rangle \equiv \sum_{M_1,M_2=-N/2}^{N/2} c^{jm}_{N/2M_1N/2M_2} |D_N^{(N/2-M_1)}\rangle \otimes |D_N^{(N/2-M_2)}\rangle , \tag{36}$$

where we use the Clebsch-Gordan coefficients $c^{jm}_{j_1m_1j_2m_2}$. Let us emphasize that the operators $\mathcal{P}_j$ do not depend on L. We can define the new positive coefficients of symmetric multiqubit states $p_j \equiv \langle\Psi| \mathcal{P}_j |\Psi\rangle$. Moreover, due to the property of the Clebsch-Gordan coefficients $c^{jm}_{j_1m_1j_2m_2} = (-1)^{j_1+j_2-j} c^{jm}_{j_2m_2j_1m_1}$, we obtain that $p_j = 0$ for $j \not\equiv N (\text{mod } 2)$. The p_j coefficients, which are linear combinations of the sector legnths S_m, fulfill the new inequalities

$$0 \leqslant p_j(S_m) \leqslant \frac{1 + (-1)^{N-j}}{2} , \tag{37}$$

for any pure symmetric states. Thus, the coefficients p_j and a_L, seen as linear functions of the S_m's, impose new numerical ranges when one is restricted to the set of symmetric states. We define this region as

$$\begin{aligned} \mathcal{R}^{\text{sym}} \equiv \mathcal{R} \cap \Big\{ \mathbf{S} \,\Big|\, & \text{Eq. (37) and } a_j(S_m) \geqslant 0 \\ & \text{holds } \forall\, j = 0, \ldots, N \Big\}. \end{aligned} \tag{38}$$

[1] The states $|\phi_{jm}^{\text{2-copy}}\rangle$ come from the coupling of two angular momentum states of spin $N/2$ with a total amplitude j [13].

To obtain the explicit expression of p_j in terms of S_m's, we first write them in terms of the a_L's

$$p_j = (-1)^{N+j}(2j+1) \sum_{L=0}^{N} \begin{Bmatrix} N/2 & N/2 & L \\ N/2 & N/2 & j \end{Bmatrix} a_L. \tag{39}$$

Since $\mathcal{R}^{\text{sym}}$ is defined by only linear inequalities, it is always a polytope on $\mathbb{R}_+^N$.

B. Other infinite families of inequalities

The extrema below are taken over normalized pure symmetric N-qubit states. We define $N_0'(m)$ and, for even m, $N_0(m)$ as the *least eventual thresholds*: the corresponding extremal equality holds for every integer N at or above the threshold.

Proposition 1 (Maximum thresholds) *For every positive integer m, the threshold $N_0'(m)$ is finite and*

$$\max_{|\psi\rangle} S_m(\psi) = \binom{N}{m} \qquad \text{for every } N \geq N_0'(m). \tag{40}$$

Coherent product states attain this maximum. The thresholds obey

$$N_0'(p+q) \leq N_0'(p) + \binom{p+2}{2} N_0'(q), \qquad p, q \geq 1, \tag{41}$$

and, for every $m \geq 3$,

$$2m - 1 \leq N_0'(m) \leq \binom{m+2}{3} - 5. \tag{42}$$

Consequently, $N_0'(m)$ grows at least linearly and at most cubically with m, with the explicit upper estimate $N_0'(m) \leq m^3/6 + O(m^2)$.[2] The certified finite-rank values and bounds for $N_0'(m)$ are given in Table I.

Proof sketch. The existence of an eventual upper bound and the small-rank bounds through $m = 4$ follow from Ref. [12]. For the lower bound, the symmetric W state satisfies

$$\frac{S_m(W_N)}{\binom{N}{m}} = 1 + \frac{4m(2m-1-N)}{N^2}, \tag{43}$$

which exceeds one at $N = 2m-2$ for $m \geq 2$. Permutation symmetry leaves $\binom{p+2}{2}$ distinct order-p Pauli correlation types. Assigning a disjoint order-q block to each type and applying the operator norm bounds for the two sectors gives (41). Setting $p = k$, $q = 1$ and iterating from $N_0'(3) = 5$ gives (42). At $m = 5$, an exact nine-qubit

[2] The notation $f(m) = O(g(m))$ means that $|f(m)| \leq C|g(m)|$ for some constant $C > 0$, independent of m, and all sufficiently large m.

counterexample and a ten-qubit two-copy spectral certificate establish the stated value. For odd $7 \le m \le 51$, exact two- or three-copy positivity certificates prove the upper bound at $N = 2m-1$. Convexity extends each bound to mixed symmetric states, and symmetric reduction extends it to every larger N. Full derivations and the exact verification data are provided in Ref. [?].

Proposition 2 (Minimum thresholds) *For every positive even integer m and every symmetric state with $N \ge m$,*

$$S_m \ge \frac{1}{m+1}\binom{N}{m}, \tag{44}$$

with equality if and only if $a_2 = a_4 = \cdots = a_m = 0$. The least eventual threshold $N_0(m)$ is finite, so that

$$\min_{|\psi\rangle} S_m(\psi) = \frac{1}{m+1}\binom{N}{m} \quad \text{for every } N \ge N_0(m). \tag{45}$$

It is nondecreasing along the even integers. There is an absolute constant $C > 0$, independent of m, such that for every even $m \ge 6$,

$$\max\left\{m+2, \frac{3m}{2}+1, \frac{m(m+6)}{8}+2\right\} \le N_0(m) \le Cm^2. \tag{46}$$

Thus, along the even integers, $N_0(m)$ is bounded above and below by positive constant multiples of m^2. The constant C in the upper bound is nonexplicit. The certified finite-rank values and bounds for $N_0(m)$ are given in Table I.

Proof sketch. Equation (23) has strictly positive coefficients at $L = 0, 2, \ldots, m$ and $a_0 = 1/(N+1)$, giving the lower bound and its equality criterion. The nested equality conditions imply monotonicity of $N_0(m)$. Eventual attainment follows from pure anticoherent-state constructions [14]. For the quantitative upper bound, take a well-separated spherical $3m$-design with $O(m^2)$ points. A localized harmonic kernel corrects the weights of a superposition of the associated spin coherent states. The correction is a contraction uniformly for $N \ge Cm^2$, yielding a normalized pure state with all multipoles of ranks $1, \ldots, m$ equal to zero at every such N. For the quadratic lower bound, time-reversal averaging converts an attaining pure state into an $(m+1)$-anticoherent state of rank at most two. Rank bounds for its tensor-operator Gram matrix, together with exclusion of the limiting rank configuration, give the last term in (46); the remaining terms follow from reduced-state rank and purity constraints. The exact small-rank thresholds combine exclusions at $N = 3, 9, 16, 24$ with attaining constructions for every larger integer. The analytic arguments, rational exclusion certificates, and exact attainment verifiers are given in Ref. [?].

Table I collects the exact values and the further certified intervals and lower bounds. Its even-rank upper bounds for $N_0'(m)$ follow from (41) with $p = 1$.

The even-sector lower bound is also attained at the individual sizes $(m, N) = (10, 34), (12, 48), (14, 62), (16, 78)$, as certified by exact root-isolation data [?]. These attainment statements do not determine the corresponding eventual thresholds.

For odd $N \geqslant 7$, the state

$$|\psi\rangle = \mathcal{N}\left(\sqrt{\frac{N+3}{2}}|D_N^{(0)}\rangle + \sqrt{N}|D_N^{((N+1)/2)}\rangle\right) \tag{47}$$

has sector length $S_2 = \binom{N}{2}/3$. For even $N \geqslant 8$, the state

$$|\psi\rangle = \mathcal{N}\left(\sqrt{\frac{N(N+2)}{2(N-1)}}|D_N^{(0)}\rangle + \sqrt{N}|D_N^{(N/2)}\rangle\right) \tag{48}$$

has sector length $S_2 = \binom{N}{2}/3$.

C. Conjectures/theorems on discrete laplacian

V. SMALL NUMBER OF QUBITS

In this section, we study the sets $\mathcal{R}^{\mathrm{sym}}$ and $\mathcal{S}^{\mathrm{sym}}$ for a small number of qubits $N \leqslant 6$, and their comparison with the non-symmetric case. We also determine the exact range of S_8 for pure symmetric states with $N = 9$. For $N \leqslant 4$, $\mathcal{R} = \mathcal{S}$ which was already proved in Ref. [12]. The symmetric-state families constructed there cover the entire allowed range, including its interior (see in particular Eqs. (47)–(50) for $N = 4$). Thus $\mathcal{S}^{\mathrm{sym}} = \mathcal{S} = \mathcal{R} = \mathcal{R}^{\mathrm{sym}}$. Attainability of the vertices alone would establish equality of convex hulls, but would not suffice to establish equality of the full numerical ranges.

A. $N = 5$

The discrepancy of $\mathcal{S}$ and $\mathcal{S}^{\mathrm{sym}}$ starts at $N = 5$. For non-symmetric states we showed that $\mathcal{S} = \mathcal{R}$ [12], where $\mathcal{R}$ is defined by the set of equalities and inequalities

$$\begin{aligned} S_3 &= 10 + 2S_1 - S_2, \\ S_4 &= 15 - S_2, \\ S_5 &= 6 - 3S_1 + S_2, \end{aligned} \tag{49}$$

$$0 \leqslant S_1 \leqslant 5, \quad 0 \leqslant S_2 \leqslant 10, \quad 2S_1 \leqslant S_2. \tag{50}$$

The set $\mathcal{R}^{\mathrm{sym}}$ for symmetric states is further restricted by new equalities, one of which, $a_2 \geqslant 0$, produces the additional more stringent condition $S_2 \geqslant \frac{10}{3}$, so that the set of inequalities (50) is replaced by

$$0 \leqslant S_1 \leqslant 5, \quad \frac{10}{3} \leqslant S_2 \leqslant 10, \quad 2S_1 \leqslant S_2. \tag{51}$$

Numerical calculation shows that the domain $\mathcal{S}^{\mathrm{sym}}$ is given by the shaded region of Fig. 2, with extremal vertices at coordinates $(0, 18/5)$ and $(9/5, 18/5)$. In the

Threshold values and small-rank bounds

m	$N_0'(m)$	$N_0(m)$
1	1	—
2	3	4
3	5	—
4	$[7, 9]$	10
5	10	—
6	$[11, 31]$	17
7	13	—
8	$[15, 40]$	25
9	17	—
10	$[19, 52]$	$\geqslant 28$
11	21	—
12	$[23, 64]$	$\geqslant 35$
13	25	—
14	$[27, 76]$	$\geqslant 46$
m odd $15 \leqslant m \leqslant 51$	$2m-1$	—

Further certified lower bounds

m	$N_0(m) \geqslant$	m	$N_0(m) \geqslant$	m	$N_0(m) \geqslant$
16	55	38	240	60	549
18	68	40	259	62	588
20	77	42	288	64	617
22	90	44	309	66	662
24	101	46	338	68	697
26	118	48	363	70	740
28	133	50	388	72	775
30	152	52	419	80	947
32	173	54	448	110	1762
34	196	56	481	120	2070
36	215	58	514	160	3561

TABLE I. Certified information on the least eventual thresholds for pure symmetric states [**?**]. In the left panel, a number without an inequality is an exact threshold; $[a, b]$ means $a \leqslant N_0'(m) \leqslant b$. A dash means that the even-sector minimum threshold is not defined for odd m. The right panel contains lower bounds, not exact thresholds. The equality $N_0'(m) = 2m-1$ is proved for every odd $7 \leqslant m \leqslant 51$ by exact two- or three-copy certificates; no assertion for arbitrary odd m is implied. The certified minimum thresholds are $N_0(2) = 4$, $N_0(4) = 10$, $N_0(6) = 17$, and $N_0(8) = 25$.

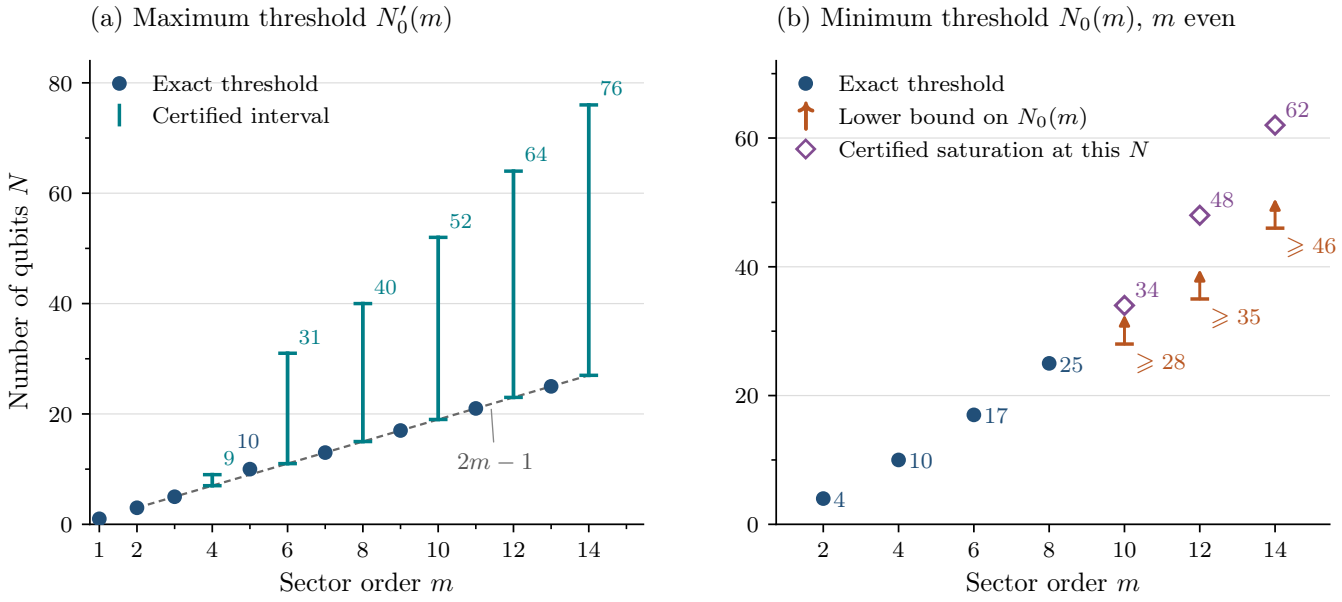


FIG. 1. Thresholds from Table I: (a) $N_0'(m)$; (b) $N_0(m)$ for even m. Filled circles denote exact thresholds, capped segments certified intervals, and upward arrows certified lower bounds. The dashed line is $2m-1$. Open diamonds certify saturation $S_m = \binom{N}{m}/(m+1)$ at the indicated individual sizes N. They do not assert saturation for every larger size and hence give no upper bound on $N_0(m)$. For odd $m < N$, $\min_{|\psi\rangle} S_m = 0$ (GHZ state), omitted from (b).

Dicke basis, symmetric states having these sector lengths are

$$\frac{1}{4}(0, 3, 0, \sqrt{2}, 0, -\sqrt{5}), \quad (0, 1, 0, 0, 0, 0), \tag{52}$$

respectively.

Numerics suggest the sharper constraint $S_2 \geqslant 18/5$ at $S_1 = 0$. A proof using Schmidt isospectrality of ρ_2, ρ_3 and their spin-multipole decompositions remains to be completed.

In order to find an analytical description of the other boundaries of $\mathcal{S}^{\mathrm{sym}}$, the strategy is as follows. First, we make use of the Majorana representation for spin states [15], where a pure symmetric N-qubit state is represented by N points on the Bloch sphere. Motivated by the numerical boundary, we investigate states with C_2 symme-

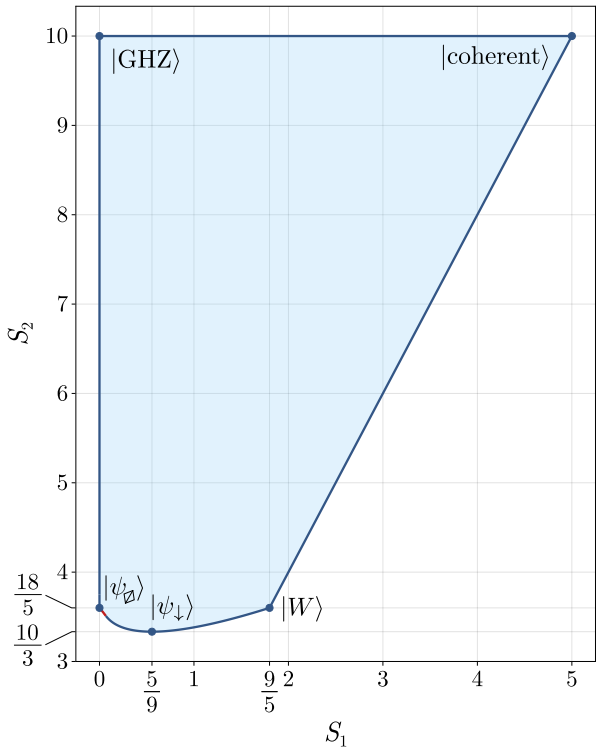


FIG. 2. Boundary

try as candidates for extrema of S_2 at fixed S_1. That is, the set of N points representing such a candidate is invariant under rotation by an angle π around some axis. Suppose the state can be written $|\psi\rangle = \sum_{i=0}^{N} c_i \, |D_N^{(i)}\rangle$ in the Dicke basis. As shown in [14], a C_2-symmetric state can be brought by some symmetric local unitary operator to a state with $c_{2i} = 0$, $0 \leqslant i \leqslant N/2$, in the Dicke basis. Appendix L derives necessary stationarity conditions supporting this restriction in a nondegenerate branch, but does not establish it for every extremizer. The C_2 symmetry reduces the number of free parameters in the state by half; the resulting analytic candidate curves are given in Appendix A. Their global optimality requires a separate argument for the exceptional branches.

B. $N = 6$

For $N = 6$, the equalities defining the numerical range $\mathcal{R}^{\mathrm{sym}}$ are

$$\begin{aligned} S_4 &= 45 + 10S_1 - 2S_2 - 3S_3, \\ S_5 &= 3S_3 - 9S_1, \\ S_6 &= 18 - 2S_1 + S_2 - S_3. \end{aligned} \tag{53}$$

and inequalities [12]

$$\begin{aligned} &0 \leqslant 10S_1 \leqslant 3S_3 \leqslant 6S_1 + 24, \quad 5 \leqslant S_2 \leqslant 15, \\ &5S_1 + 5 \leqslant S_2 + S_3, \quad 10S_2 + 3S_3 - 162 \leqslant 8S_1, \\ &20S_2 + 21S_3 \leqslant 70S_1 + 324, \end{aligned} \tag{54}$$

TABLE II. Sector lengths (S_1), (S_1, S_2), and (S_1, S_2, S_3) of selected pure symmetric states. For $N \leqslant 4$, the listed points are vertices of the attainable range. For $N = 5, 6$, the table also includes extrema and distinguished boundary points; it does not identify the attainable range with the linear outer polytope. The state definitions are given in Appendix A.

N	State	(S_1) or (S_1, S_2)
2	$\|\mathrm{GHZ}(2)\rangle$	(0)
	$\|0\rangle^{\otimes 2}$	(2)
3	$\|\mathrm{GHZ}(3)\rangle$	(0)
	$\|0\rangle^{\otimes 3}$	(3)
4	$\|\mathrm{tetra}\rangle$	$(0, 2)$
	$\|\mathrm{GHZ}(4)\rangle$	$(0, 6)$
	$\|0\rangle^{\otimes 4}$	$(4, 6)$
5	$\|\psi_5^{(a)}\rangle$	$(5/9, 10/3)$
	$\|\psi_5^{(b)}\rangle$	$(0, 18/5)$
	$\|D_5^{(1)}\rangle$	$(9/5, 18/5)$
	$\|\mathrm{GHZ}(5)\rangle$	$(0, 10)$
	$\|0\rangle^{\otimes 5}$	$(5, 10)$
N	State	(S_1, S_2, S_3)
6	$\|\mathrm{octa}\rangle$	$(0, 5, 0)$
	$\|\mathrm{GHZ}(6)\rangle$	$(0, 15, 0)$
	$\|D_6^{(1)}\rangle$	$(8/3, 5, 40/3)$
	$\|0\rangle^{\otimes 6}$	$(6, 15, 20)$
	$\|\psi_6\rangle$, $\|\chi_6\rangle$	$(24, 405, 440)/49$
	$\|\mathrm{AC}_2\rangle$	$(0, 5, 640/81)$
	$\|\psi_0\rangle$	$(0, 51/10, 8)$
	$\|\mathrm{pyramid}\rangle$	$(0, 27/5, 8)$
	$\|\Xi_*\rangle$	$(0, s_*, 8)$

Here $s_* = 13 - 8\sqrt{15}/5$. The first four six-qubit rows are attained vertices of the linear outer polytope; the remaining rows are attained extrema or boundary points. The table does not enumerate all boundary points.

I haven't checked if there is a more compact expression for these inequalities.

To derive the new inequalities, both sets of inequalities $a_L \geqslant 0$ and $p_J \geqslant 0$ are complementary.

These inequalities restrict the values of S_m's to the separate intervals $S_1 \in [0, 6]$, $S_2 \in [5, 15]$, $S_3 \in [0, 20]$ and $S_4 \in \left[\frac{27}{5}, 35\right]$, whereas the pure-state optimization gives the tighter candidate $S_4 \in \left[\frac{45}{7}, 35\right]$ with state (A25) reaching the lower bound.

Table II collects the independent sector coordinates of selected pure symmetric states for $2 \leqslant N \leqslant 6$. Their representatives are defined in Appendix A.

C. Sharp eighth-sector bounds for nine qubits

The eighth sector provides an example for which both extrema can be determined exactly on the full complex symmetric pure-state space.

Theorem 1 *Every normalized pure state in* $\mathrm{Sym}^9(\mathbb{C}^2)$ *satisfies*

$$9 \leqslant S_8 \leqslant \nu_9, \qquad \nu_9 = \frac{35104185 + 770688\sqrt{669}}{345193}. \quad (55)$$

Both bounds are attained. Every generalized GHZ state

$$|g_{\alpha,\beta}\rangle = \alpha\,|D_9^{(0)}\rangle + \beta\,|D_9^{(9)}\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \quad (56)$$

attains the minimum. A global maximizer is

$$|\psi_9^{(e)}\rangle = \frac{|D_9^{(1)}\rangle + \sqrt{14u}\,|D_9^{(4)}\rangle + 2t\,|D_9^{(7)}\rangle}{\sqrt{1+14u+4t^2}}, \quad (57)$$

where

$$t = \frac{83 + 9\sqrt{669}}{430}, \qquad u = \frac{37052 + 171\sqrt{669}}{210700}. \quad (58)$$

Proof. Appendix B sketches the exact sum-of-squares proof of both inequalities for arbitrary complex Dicke coefficients. For the states in Eq. (56), every eight-qubit reduction is a mixture of the all-zero and all-one states. Its only nonzero weight-eight Pauli expectation is $\langle\sigma_3^{\otimes 8}\rangle = 1$. Summing over the nine choices of omitted qubit gives $S_8 = 9$. Substitution of Eqs. (57) and (58) into the reduced-purity formula (B1) gives $S_8 = \nu_9$ exactly. Thus the certified bounds are sharp. □

Numerically, $\nu_9 = 159.441403981044\ldots$, and the three nonzero Dicke amplitudes of $|\psi_9^{(e)}\rangle$ are approximately 0.411237517885, 0.682680129629, and 0.604012867821, in increasing excitation number. These decimal values are only illustrations; both the bound and its proof use exact arithmetic.

Collective rotations $U^{\otimes 9}$ of any equality state are again equality states. There is a further continuous family: if $T\,|\psi\rangle = |\tilde{\psi}\rangle$ denotes the time reversal in Eq. (18), then

$$|\psi_{\alpha,\beta}\rangle = \alpha\,|\psi\rangle + \beta T\,|\psi\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \quad (59)$$

has the same even-weight Pauli expectations as $|\psi\rangle$. Indeed, $T^2 = -1$, so $|\psi\rangle$ and $T\,|\psi\rangle$ are orthogonal. Every even-weight Pauli operator commutes with T, and its compression to this two-dimensional subspace is proportional to the identity. Equation (59) therefore preserves S_8, yielding families of both minimizers and maximizers. We do not assert that these families exhaust all equality states.

VI. EXTENSIONS AND CONJECTURES: MIXED STATES,...

VII. CONCLUSIONS

ACKNOWLEDGEMENTS

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Appendix A: Optimal states

In this appendix and Table III, extrema and minimal thresholds without a supplied proof or supporting reference are understood as conjectural. Evaluating a representative establishes attainability; global optimality requires a bound. Eventual attainability of the even-sector minimum is proved, but specific threshold values must be distinguished from its existence; the certified information is summarized in Table I. In this Appendix, we present analytical expressions for proven or putative optimal states that extremize certain sector lengths.

Common representatives

For Tables II and III, we use the following normalized representatives in the Dicke basis:

$$|\mathrm{GHZ}(N)\rangle = \frac{|D_N^{(0)}\rangle + |D_N^{(N)}\rangle}{\sqrt{2}},$$
$$|\mathrm{tetra}\rangle = \frac{|D_4^{(0)}\rangle + \sqrt{2}\,|D_4^{(3)}\rangle}{\sqrt{3}},$$
$$|\mathrm{octa}\rangle = \frac{|D_6^{(1)}\rangle + |D_6^{(5)}\rangle}{\sqrt{2}}. \quad (A1)$$

The coherent state is $|0\rangle^{\otimes N} = |D_N^{(0)}\rangle$.

$N = 5$

- The five-qubit state

$$|\psi_5^{(a)}\rangle = \sqrt{\tfrac{5}{6}}\,|D_5^{(1)}\rangle + \sqrt{\tfrac{1}{6}}\,|D_5^{(5)}\rangle \quad (A2)$$

minimizes S_2. It has sector lengths

$$\mathbf{S} = \left(\frac{5}{9}, \frac{10}{3}, \frac{70}{9}, \frac{35}{3}, \frac{23}{3}\right). \quad (A3)$$

Its one-qubit reduced states have eigenvalues $\left(\frac{1}{3}, \frac{2}{3}\right)$, while its two-qubit reduced states have non-zero eigenvalues $\left(\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\right)$.

- The five-qubit state

$$|\psi_5^{(b)}\rangle = \frac{1}{4}\left(\sqrt{5}\,|D_5^{(0)}\rangle + \sqrt{2}\,|D_5^{(2)}\rangle - 3\,|D_5^{(4)}\rangle\right). \quad (A4)$$

minimizes S_1 and has sector lengths

$$\mathbf{S} = \left(0, \frac{18}{5}, \frac{32}{5}, \frac{57}{5}, \frac{48}{5}\right). \quad (A5)$$

- The five-qubit state

$$|\psi_5^{(c)}\rangle = \frac{1}{\sqrt{218}}\left(\sqrt{5\left(17+6\sqrt{5}\right)}\,|D_5^{(1)}\rangle + \sqrt{133-30\sqrt{5}}\,|D_5^{(5)}\rangle\right). \quad (A6)$$

has C_4 symmetry. It is an inverse square pyramid with polar angle base $\theta = 2\tan^{-1}\left(\sqrt[4]{\frac{2}{\sqrt{5}}-\frac{3}{5}}\right)$. It has sector lengths $S_1 = \frac{22805-9840\sqrt{5}}{11881}$ and $S_2 = \frac{10\left(7283-1392\sqrt{5}\right)}{11881}$.

Boundary:

We consider symmetric five-qubit states written in the Dicke basis $|\psi\rangle = \sum_{k=0}^{5} d_k |D_5^{(k)}\rangle$, with Dicke vector $\mathbf{d} = (d_0, d_1, d_2, d_3, d_4, d_5)$. A C_2-symmetric family can be parametrized as

$$\mathbf{d}_{C_2}(t) = \frac{1}{2\sqrt{t^2-1}}\left(1, 0, \sqrt{3t^2-5}, 0, t, 0\right). \tag{A7}$$

The parameter takes values in the two intervals

$$t \in \left[\frac{\sqrt{15}}{3}, \sqrt{5}\right] \cup \left[-\sqrt{\frac{345+60\sqrt{15}}{289}}, -\frac{3}{\sqrt{5}}\right]. \tag{A8}$$

For this family, the sector lengths satisfy

$$S_1 = 0, \tag{A9}$$

and

$$S_2(t) = \frac{t\left[t\left(177t^2+36\sqrt{5}\,t-410\right)-60\sqrt{5}\right]+225}{20(t^2-1)^2}. \tag{A10}$$

The negative branch gives $S_2 \in \left[\frac{18}{5}, \frac{15}{4}\right]$, whereas the positive branch gives $S_2 \in \left[\frac{15}{4}, 10\right]$. Thus, the combined family covers $S_2 \in \left[\frac{18}{5}, 10\right]$.

We also consider the one-parameter GHZ family

$$|\mathrm{GHZ}(5,\phi)\rangle = \cos(\tfrac{\phi}{2})\,|D_5^{(0)}\rangle + \sin(\tfrac{\phi}{2})\,|D_5^{(5)}\rangle, \tag{A11}$$

with $0 \leqslant \phi \leqslant \frac{\pi}{2}$. It has sector lengths $S_1 = 5\cos^2\phi \in [0,5]$ with $S_2 = 10$.

Another family is

$$\mathbf{d}_\infty(\lambda) = \left(0, \frac{1}{\sqrt{1+\lambda^2}}, 0, 0, 0, \frac{\lambda}{\sqrt{1+\lambda^2}}\right), \tag{A12}$$

with

$$0 \leqslant \lambda \leqslant 2 - \frac{3}{\sqrt{5}}. \tag{A13}$$

In the limit $\lambda \to 0$, this family tends to $|D_5^{(1)}\rangle$. It has sector lengths $S_1 \in \left[\frac{22805-9840\sqrt{5}}{11881}, \frac{9}{5}\right]$ and $S_2 \in \left[\frac{10}{3}, \frac{18}{5}\right]$ given by

$$S_1 = \frac{\left(3-5\lambda^2\right)^2}{5\left(\lambda^2+1\right)^2}, \quad S_2 = \frac{2\left(25\lambda^4+10\lambda^2+9\right)}{5\left(\lambda^2+1\right)^2}, \tag{A14}$$

so that

$$S_2 = \frac{1}{4}\left(3S_1 - 2\sqrt{5}\sqrt{S_1} + 15\right) \tag{A15}$$

Finally, we consider the family

$$\mathbf{d}(\theta) = (\cos\theta, \sin\theta, 0, 0, 0, 0), \tag{A16}$$

with $0 \leqslant \theta \leqslant \frac{\pi}{2}$. It has sector lengths $S_1 \in \left[\frac{9}{5}, 5\right]$ and $S_2 \in \left[\frac{18}{5}, 10\right]$ given by

$$S_1 = 5 - \frac{16}{5}\sin^4\theta, \qquad S_2 = 2S_1. \tag{A17}$$

The remaining numerically observed boundary segment is reproduced by the following family and its explicit parametrization. The family corresponds to $S_1 \in [0, \frac{22805-9840\sqrt{5}}{11881}]$ and is of the form

$$\mathcal{N}(-p, 0, q, 0, 1, 0), \tag{A18}$$

with sector lengths

$$S_1 = \frac{\left(5p^2+q^2-3\right)^2}{5\left(p^2+q^2+1\right)^2},$$
$$S_2 = \frac{2\left(25p^4+10p^2\left(q^2+1\right)-24\sqrt{5}pq^2+19q^4+58q^2+9\right)}{5\left(p^2+q^2+1\right)^2}. \tag{A19}$$

A convenient parametrization of the candidate boundary is obtained by imposing the fixed-S_1 constraint directly. Let

$$u(p) = -\frac{15\sqrt{5}p^4+20p^3+18\sqrt{5}p^2-28p-9\sqrt{5}}{3\sqrt{5}p^4+12p^3+18\sqrt{5}p^2-20p+3\sqrt{5}}, \tag{A20}$$

with

$$2 - \frac{3}{\sqrt{5}} \leq p \leq \frac{\sqrt{5}}{3}. \tag{A21}$$

Then the corresponding value of q is determined by

$$q^2(p) = \frac{2p\left(5p^2+6\sqrt{5}p-11\right)}{3\sqrt{5}p^2+2p+3\sqrt{5}}. \tag{A22}$$

The candidate curve is therefore parametrized as

$$S_1(p) = \frac{u(p)^2}{5}, \tag{A23}$$

and

$$S_2(p) = \frac{2\left(1125p^8+600\sqrt{5}p^7+3100p^6+1800\sqrt{5}p^5+22190p^4-14424\sqrt{5}p^3+26300p^2-3720\sqrt{5}p+405\right)}{5\left(45p^8+72\sqrt{5}p^7+684p^6+312\sqrt{5}p^5+1230p^4-648\sqrt{5}p^3+940p^2-120\sqrt{5}p+45\right)} \tag{A24}$$

These formulas give a single explicit parametrization of this boundary segment.

$N = 6$

- The symmetric six-qubit state

$$|\psi_6\rangle = \mathcal{N}\Big(|D_6^{(5)}\rangle + \sqrt{20}\,|D_6^{(2)}\rangle\Big) \tag{A25}$$

minimizes S_4. It has sector lengths

$$\mathbf{S} = \left(\frac{24}{49}, \frac{405}{49}, \frac{440}{49}, \frac{45}{7}, \frac{1104}{49}, \frac{799}{49}\right). \tag{A26}$$

- A 1-anticoherent symmetric six-qubit state is given by

$$|\psi\rangle = -\tfrac{1}{2}\,|D_6^{(0)}\rangle + \tfrac{\sqrt{3}}{2}\,|D_6^{(4)}\rangle\,. \tag{A27}$$

Its Majorana representation consists of points located at the vertices of a square pyramid, with the apex being twice degenerate. The base of the pyramid lies at polar angle

$$\theta = 2\arctan\left(3^{1/4}5^{1/8}\right). \tag{A28}$$

This state minimizes S_1 and maximizes S_5. It has sector lengths

$$\mathbf{S} = \left(0, \frac{27}{5}, 8, \frac{51}{5}, 24, \frac{77}{5}\right). \tag{A29}$$

The state in Eq. (A27) is denoted $|\text{pyramid}\rangle$ in Table II. The additional six-qubit representatives in that table are

$$|\chi_6\rangle = \frac{2\,|D_6^{(0)}\rangle + \sqrt{3}\,|D_6^{(5)}\rangle}{\sqrt{7}}, \tag{A30}$$

$$|\mathrm{AC}_2\rangle = \frac{\sqrt{2}\,|D_6^{(0)}\rangle + \sqrt{5}\,|D_6^{(3)}\rangle + \sqrt{2}\,|D_6^{(6)}\rangle}{3}, \tag{A31}$$

$$|\psi_0\rangle = \frac{|D_6^{(0)}\rangle + \sqrt{2}\,|D_6^{(3)}\rangle + |D_6^{(6)}\rangle}{2}. \tag{A32}$$

In particular, $|\chi_6\rangle$ and $|\psi_6\rangle$ in Eq. (A25) have the same sector lengths. Finally, put $u_* = (\sqrt{5}-\sqrt{3})/4$ and define

$$\begin{aligned}|\Xi_*\rangle = \tfrac{1}{2}\big[&\sqrt{(1+u_*)(1-2u_*)}\,|D_6^{(0)}\rangle \\ &+ \sqrt{(1+u_*)(1+2u_*)}\,|D_6^{(2)}\rangle \\ &+ \sqrt{(1-u_*)(1-2u_*)}\,|D_6^{(4)}\rangle \\ &- \sqrt{(1-u_*)(1+2u_*)}\,|D_6^{(6)}\rangle\big].\end{aligned} \tag{A33}$$

This normalized state has $(S_1, S_2, S_3) = (0, s_*, 8)$, where $s_* = 13 - 8\sqrt{15}/5$. The coordinates in Table II follow by evaluating reduced purities and using their relations to the sector lengths.

$N = 7$

- The symmetric seven-qubit state

$$|\psi_7\rangle = \mathcal{N}\Big(2\sqrt{2}\,|D_7^{(0)}\rangle + \sqrt{14}\,|D_7^{(3)}\rangle + \sqrt{14}\,|D_7^{(6)}\rangle\Big) \tag{A34}$$

is a pure symmetric state with maximally mixed single-qubit reduced density matrices. All two-qubit reduced states have non-zero eigenvalues $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$, while all three-qubit reduced states have non-zero eigenvalues $\left(\frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10}\right)$. This state is 2-anticoherent and realizes extremal values of S_m for $m = 4$ and $m = 6$. It has sector lengths

$$\mathbf{S} = (0, 7, 14, 7, 28, 49, 22)\,. \tag{A35}$$

- The symmetric seven-qubit state

$$|\psi_7^{(b)}\rangle = \mathcal{N}\Big(0, -1, 0, 0, \sqrt{10}, 0, 0, \sqrt{7}\Big) \tag{A36}$$

maximizes S_4 and has sector lengths

$$\mathbf{S} = \left(\frac{9}{7}, 7, 5, \frac{265}{7}, \frac{223}{7}, \frac{127}{7}, \frac{181}{7}\right). \tag{A37}$$

$N = 8$

- The 2-anticoherent 8-qubit symmetric state

$$|\text{tetra}(8)\rangle = \frac{1}{3\sqrt{3}}\left(\sqrt{7}\,|D_8^{(0)}\rangle + 2\,|D_8^{(3)}\rangle + 4\,|D_8^{(6)}\rangle\right) \tag{A38}$$

minimizes both S_1 and maximizes S_7 (numerical observation). Its sector lengths are equal to

$$\mathbf{S} = \left(0, \frac{28}{3}, \frac{144}{7}, \frac{310}{21}, \frac{160}{7}, \frac{1396}{21}, \frac{592}{7}, \frac{255}{7}\right). \tag{A39}$$

- The symmetric eight-qubit state $|\psi_8\rangle$ is given in the Dicke basis by

$$|\psi_8\rangle = \mathcal{N}\big(\sqrt{30}, 0, 0, 0, 2\sqrt{21}, 0, 0, 0, \sqrt{30}\big). \tag{A40}$$

It is a 3-anticoherent state and achieves extremal sector lengths for $m = 2$ and $m = 6$ among symmetric states. It has sector lengths

$$\mathbf{S} = \left(0, \frac{28}{3}, 0, \frac{154}{3}, 0, \frac{364}{3}, 0, 73\right). \tag{A41}$$

- The symmetric eight-qubit state

$$|\psi_8^{(b)}\rangle = \mathcal{N}\Big(0, 0, 4, 0, 0, 2, 0, 0, \sqrt{7}\Big) \tag{A42}$$

minimizes S_4 and has sector lengths

$$\mathbf{S} = \left(0, \frac{28}{3}, \frac{144}{7}, \frac{310}{21}, \frac{160}{7}, \frac{1396}{21}, \frac{592}{7}, \frac{255}{7}\right). \tag{A43}$$

$N = 9$

- The symmetric nine-qubit state

$$|\psi_9\rangle = \mathcal{N}\big(1,0,0,\sqrt{2},0,0,0,0,0,0\big) \tag{A44}$$

is expressed in the Dicke basis and appears as a numerically optimal state for $t = 2$. It has sector lengths

$$\mathbf{S} = \left(\frac{25}{9}, 12, \frac{3772}{189}, \frac{470}{9}, \frac{6938}{63}, \frac{940}{9}, \frac{5396}{63}, \frac{259}{3}, \frac{7085}{189}\right). \tag{A45}$$

- The state

$$|\psi_9^{(b)}\rangle = \mathcal{N}\Big(5,0,0,0,0,0,\sqrt{66},0,0,0\Big) \tag{A46}$$

minimizes $S_4 = 2754/91$. It has sector lengths

$$\mathbf{S} = \left(\frac{81}{8281}, \frac{100908}{8281}, \frac{1281204}{57967}, \frac{2754}{91}, \frac{3165642}{57967}, \frac{70068}{1183}, \frac{6535476}{57967}, \frac{1269657}{8281}, \frac{3856663}{57967}\right). \tag{A47}$$

- The state

$$|\psi_9^{(c)}\rangle \approx (0.8646511, 0, 0, 0.4913279, 0, 0, -0.1017967, 0, 0, -0.0247553) \tag{A48}$$

is a numerical candidate for the maximum, with $S_5 \approx 126.1380053$. Its sector lengths are approximately

$$\mathbf{S} \simeq (6.1111, 21.2957, 47.0085, 88.1911, 126.1380, 111.3121, 65.7838, 34.2011, 10.9585)\,. \tag{A49}$$

- The state $|\psi_9^{(e)}\rangle$ in Eq. (57) globally maximizes $S_8 = \nu_9$. The exact parameters, the generalized GHZ minimizers with $S_8 = 9$, and continuous families of equality states are given in Sec. V C. Their global optimality is established by Theorem 1 and Appendix B.

$N = 10$

- The symmetric ten-qubit state

$$|\psi_{10}\rangle = \mathcal{N}\big(0,1,0,0,0,0,0,1,0,0,0\big) \tag{A50}$$

is an extremal state for $t = 2$, obtained from numerical optimization within the symmetric subspace. It has sector lengths

$$\mathbf{S} = \left(\frac{2}{5}, 15, \frac{101}{15}, \frac{175}{3}, \frac{511}{5}, \frac{721}{5}, \frac{917}{5}, \frac{686}{5}, \frac{3289}{15}, \frac{2344}{15}\right). \tag{A51}$$

$$|\psi_{10}^{(b)}\rangle \simeq (0.1503232, 0, 0, 0.5933365, 0, 0, 0.7413402, 0, 0, 0.2752624, 0) \tag{A52}$$

is a numerical candidate for the minimum of S_6. Its sector lengths are approximately

$$\mathbf{S} \simeq (0.0005, 20.2333, 35.6982, 66.0444, 86.0971, 73.0463, 104.7418, 209.8735, 285.4625, 141.8026)\,. \tag{A53}$$

$$|\psi_{10}^{(c)}\rangle = \mathcal{N}\big(0,\sqrt{5},0,0,0,\sqrt{6},0,0,0,\sqrt{5},0\big) \tag{A54}$$

maximizes S_6. It has sector lengths

$$\mathbf{S} = (0, 15, 0, 90, 0, 270, 0, 405, 0, 243)\,. \tag{A55}$$

$$|\psi_{10}^{(d)}\rangle = \mathcal{N}\big(1,0,0,0,0,\sqrt{2},0,0,0,0,1\big) \tag{A56}$$

maximizes S_7. It has sector lengths

$$\mathbf{S} = \left(0, \frac{95}{6}, 0, \frac{280}{3}, 32, \frac{325}{3}, 320, 75, 160, \frac{437}{2}\right). \tag{A57}$$

$N = 11$

- The symmetric eleven-qubit state

$$|\psi_{11}\rangle = \mathcal{N}\big(0, 0, \sqrt{17}, 0, 0, 0, 0, 0, 1, 0, 0, 0\big) \tag{A58}$$

appears as a numerically optimal symmetric state for $t = 2$. It has sector lengths

$$\mathbf{S} = \left(\frac{361}{99}, \frac{55}{3}, \frac{12985}{243}, \frac{364690}{2673}, \frac{413126}{2673}, \frac{241766}{891}, \frac{3690}{11}, \frac{413695}{891}, \frac{91445}{243}, \frac{354401}{2673}, \frac{268879}{2673}\right). \tag{A59}$$

$N = 12$

- The symmetric twelve-qubit state

$$|\psi_{12}\rangle = \mathcal{N}\big(0, 0, \sqrt{7}, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0\big) \tag{A60}$$

is expressed in the Dicke basis and realizes extremal values of S_2 among symmetric twelve-qubit states. It has sector lengths

$$\mathbf{S} = \left(\frac{49}{12}, 22, \frac{8575}{132}, \frac{2020}{11}, \frac{6419}{22}, \frac{3848}{11}, \frac{1197}{2}, \frac{9665}{11}, \frac{2835}{4}, \frac{16766}{33}, \frac{15309}{44}, \frac{4516}{33}\right). \tag{A61}$$

Appendix B: Exact certificates for the nine-qubit eighth sector

We sketch the computer-assisted proof of Theorem 1. The explicit certificates and independent exact-arithmetic verifiers are available in the GitHub repository of Ref. [22].

For an unnormalized symmetric vector $|c\rangle = \sum_{k=0}^{9} c_k |D_9^{(k)}\rangle$, set $c_k = \sqrt{\binom{9}{k}}\, z_k$ and $n = \langle c|c\rangle$. Let $P_8(z,\bar{z})$ be its homogeneous quartic sector polynomial, so that the normalized sector length is $S_8 = P_8/n^2$. It can be computed from the unnormalized reductions $\rho_r = \mathrm{Tr}_{9-r} |c\rangle\langle c|$ as

$$P_8 = 9\sum_{r=0}^{8}(-1)^{8-r}\binom{8}{r}2^r \,\mathrm{Tr}(\rho_r^2), \tag{B1}$$

with $\rho_0 = (n)$.

The proof consists of two exact sum-of-squares identities,

$$n(P_8 - 9n^2) = \sum_a \mathbf{p}_a^\dagger G_a \mathbf{p}_a, \tag{B2}$$

$$n^2(\nu_9 n^2 - P_8) = \sum_b \mathbf{h}_b^\dagger B_b \mathbf{h}_b. \tag{B3}$$

Here the polynomial vectors $\mathbf{p}_a$ and $\mathbf{h}_b$ are homogeneous of total degree three and four, respectively, in $z, \bar{z}$. The Gram matrices G_a have rational entries, and the B_b have entries in $\mathbb{Q}(\sqrt{669})$. All are positive definite, as verified by exact symmetric elimination. The first identity is checked coefficient by coefficient; the second is checked by injective exact interpolation in the complete 88-dimensional space of real-coefficient Hermitian SU(2)-invariant polynomials of bidegree $(4,4)$. Thus the identities hold for arbitrary complex coefficients, without numerical tolerances or a restriction to a state ansatz.

For $n > 0$, dividing the two inequalities by n^3 and n^4, respectively, gives $9 \leqslant S_8 \leqslant \nu_9$. The generalized GHZ states in Eq. (56) attain the lower bound. Direct substitution of Eqs. (57) and (58) into Eq. (B1) gives $P_8 = \nu_9 n^2$, proving that the upper bound is also attained.

Appendix C: About multipole operators

The multipole operators appear originally in spin-j systems.

$$T_{LM}^{j} = \sqrt{\frac{2L+1}{2j+1}} \sum_{m,m'} c_{jmLM}^{jm'} |j,m'\rangle\langle j,m|, \tag{C1}$$

where $c_{jmLM}^{jm'}$ are the Clebsch-Gordan coefficients and $|j,m\rangle$ is the spin-j eigenstate of J_z with eigenvalue m. There exists an isomorphism between spin-j states and symmetric Dicke states [11], namely $|D_N^{(\alpha)}\rangle = |j,m\rangle$ with $N = 2j$ and $\alpha = j - m$. We define the multipole tensor operators $T_{LM}^{(N)}$ via Eq. (C1) replacing the spin states with their corresponding Dicke states. Thus, the operators $T_{LM}^{(N)}$ act on the Hilbert space of N qubits $\mathcal{H}^{\otimes N}$ and they have support on the symmetric sector $\mathcal{H}^{\vee N}$. For nonnegative multiplicities summing to N, write $\widetilde{\sigma}_{(\nu_0,\nu_1,\nu_2,\nu_3)} = L_{(\mathbf{0}_{\nu_0}\mathbf{1}_{\nu_1}\mathbf{2}_{\nu_2}\mathbf{3}_{\nu_3})}$. With the convention in Eq. (C1), $[J_z, T_{LM}] = MT_{LM}$. A relation between these operators is

$$\begin{aligned} T_{LM}^{(N)} = &\sum_{k=\max(0,M)}^{\lfloor (L+M)/2\rfloor} A(N,L,M,k) \\ &\times \sum_{r=0}^{2k-M} i^{M+r} K_r(k,2k-M;2) \\ &\times \widetilde{\sigma}_{(N-L,r,2k-M-r,L+M-2k)}, \end{aligned} \tag{C2}$$

with

$$\begin{aligned} A(N,L,M,k) \equiv &\sqrt{2L+1}\sqrt{\frac{(L+M)!(L-M)!}{(N+L+1)!(N-L)!}} \\ &\times \frac{(-1)^k 2^{M-2k} N!}{k!(k-M)!(L+M-2k)!}, \end{aligned} \tag{C3}$$

and where K_r are the Kravchuk polynomial, whose general form is [23, 24] CHECK ALSO BOOK MACWILLIAMS The Theory of Error-Correcting Codes, CHAPTER 6

$$\begin{aligned} K_r(k,n;x) \equiv &\sum_{q=0}^{r} \binom{n-k}{r-q}\binom{k}{q}(-1)^q (x-1)^{r-q} \\ =&(-1)^r \sum_{q=0}^{r} \binom{n-k}{q}\binom{k}{r-q}(1-x)^q. \end{aligned} \tag{C4}$$

In particular, the generating function of the Kravchuk polynomials are

$$\big(1+(x-1)z\big)^{n-k}\big(1-z\big)^{k} = \sum_{r=0}^{n} K_r(k,n;x) z^r. \tag{C5}$$

We now prove Eq. (C2). First, we start with a result from Ref. [10]

$$\begin{aligned} T_{LM}^{(N)} = &\sum_{k=\max(0,M)}^{\lfloor (L+M)/2\rfloor} A(N,L,M,k) \\ &\times \Pi_{\mathrm{sym}} \left(\sigma_0^{\otimes\nu_0} \otimes \sigma_-^{\otimes\nu_-} \otimes \sigma_+^{\otimes\nu_+} \otimes \sigma_3^{\otimes\nu_3}\right) \Pi_{\mathrm{sym}} \end{aligned} \tag{C6}$$

with $\sigma_\pm = \sigma_1 \pm i\sigma_2$ and $(\nu_0,\nu_-,\nu_+,\nu_3) = (N-L, k-M, k, L+M-2k)$. The k-th operator of the r.h.s. of (C6)

N	$\min S_m$	optimal state	$\max S_m$	optimal state
$m=1$				
$N \geqslant 2$	0[a]	$\|\mathrm{GHZ}(N)\rangle$	N[b]	$\|0\rangle^{\otimes N}$
$m=2$				
2	1[c]	$\|0\rangle^{\otimes 2}$	3[b]	$\|\mathrm{GHZ}(2)\rangle$
3	3[b]	any $\|\psi\rangle$	3[b]	any $\|\psi\rangle$
4	2[d]	$\|\mathrm{tetra}\rangle$	6[b]	$\|0\rangle^{\otimes 4}$
$N \geqslant 4$	$\frac{1}{3}\binom{N}{2}$		$\binom{N}{2}$	$\|0\rangle^{\otimes N}$
$m=3$				
3	1[c]	$\|0\rangle^{\otimes 3}$	4[c]	$\|\mathrm{GHZ}(3)\rangle$
4	0[a]	$\|\mathrm{GHZ}(4)\rangle$	8[b]	$\|\mathrm{tetra}\rangle$
$N \geqslant 5$	0[a]	$\|\mathrm{GHZ}(N)\rangle$	$\binom{N}{3}$[b]	$\|0\rangle^{\otimes N}$
$m=4$				
4	1[c]	$\|0\rangle^{\otimes 4}$	9[b]	$\|\mathrm{GHZ}(4)\rangle$
5	5	$\|0\rangle^{\otimes 5}$	35/3	$\|\psi_5^{(a)}\rangle$
6	45/7	$\|\psi_6\rangle$	35	$\|\mathrm{octa}\rangle$
7	7	$\|\psi_7\rangle$ (A34)	265/7	$\|\psi_7^{(b)}\rangle$ (A36)
8	310/21	$\|\psi_8^{(b)}\rangle$ (A42)	70	$\|0\rangle^{\otimes 8}$
9	2754/91	$\|\psi_9^{(b)}\rangle$ (A46)	126	$\|0\rangle^{\otimes 9}$
$N \geqslant 10$	$\frac{1}{5}\binom{N}{4}$[a]		$\binom{N}{4}$[b]	$\|0\rangle^{\otimes N}$
$m=5$				
5	1[c]	$\|0\rangle^{\otimes 5}$	16[c]	$\|\mathrm{GHZ}(5)\rangle$
6	0[a]	$\|\mathrm{GHZ}(6)\rangle$	24	$\|\mathrm{pyramid}\rangle$ (A27)
7	0[a]	$\|\mathrm{GHZ}(7)\rangle$	267/7	$\|D_7^{(1)}\rangle$
8	0[a]	$\|\mathrm{GHZ}(8)\rangle$	147/2	$\|D_8^{(1)}\rangle$
9	0[a]	$\|\mathrm{GHZ}(9)\rangle$	126.138	$\|\psi_9^{(c)}\rangle$ (A48)
$N \geqslant 10$	0	$\|\mathrm{GHZ}(N)\rangle$	$\binom{N}{5}$	$\|0\rangle^{\otimes N}$
$m=6$				
6	1[c]	$\|0\rangle^{\otimes 6}$	33[b]	$\|\mathrm{GHZ}(6)\rangle$
7	7	$\|0\rangle^{\otimes 7}$	49	$\|\psi_7\rangle$ (A34)
8	177/14	$\|D_8^{(3)}\rangle$	364/3	$\|\psi_8\rangle$
9	420/11		1204/9	$\|D_9^{(1)}\rangle$
10	73.046	$\|\psi_{10}^{(b)}\rangle$ (A52)	270	$\|\psi_{10}^{(c)}\rangle$ (A54)
$N \geqslant 17$	$\frac{1}{7}\binom{N}{6}$[a]		$\binom{N}{6}$[b]	$\|0\rangle^{\otimes N}$
$m=7$				
7	1[c]	$\|0\rangle^{\otimes 7}$	64[c]	$\|\mathrm{GHZ}(7)\rangle$
8	0[a]	$\|\mathrm{GHZ}(8)\rangle$	592/7	$\|\mathrm{tetra}(8)\rangle$ (A38)
9	0	$\|\mathrm{GHZ}(9)\rangle$	1234/9	$\|D_9^{(2)}\rangle$
10	0	$\|\mathrm{GHZ}(10)\rangle$	320	$\|\psi_{10}^{(d)}\rangle$ (A56)
$N \geqslant ?$	0[a]	$\|\mathrm{GHZ}(N)\rangle$	?	?
$m=8$				
8	1[c]	$\|0\rangle^{\otimes 8}$	129[b]	$\|\mathrm{GHZ}(8)\rangle$
9	9	$\|\mathrm{GHZ}(9)\rangle$	ν_9	$\|\psi_9^{(e)}\rangle$ (57)
$m=N$				
N odd	1[c]	$\|0\rangle^{\otimes N}$	2^{N-1}[c]	$\|\mathrm{GHZ}(N)\rangle$
N even	1[c]	$\|0\rangle^{\otimes N}$	$2^{N-1}+1$[f]	$\|\mathrm{GHZ}(N)\rangle$

[a] $S_m \geqslant 0$ [b] [6] [c] [5] [d] [16, 17] [e] [18–20] [f] [21]

TABLE III. Examples of optimal N-qubit pure symmetric states. For $N=9$, $m=8$, both extrema are proved in Theorem 1; ν_9 is given in Eq. (55).

is equal to

$$\begin{aligned}
&\Pi_{\rm sym}\left(\sigma_0^{\otimes\nu_0}\otimes\sigma_-^{\otimes\nu_-}\otimes\sigma_+^{\otimes\nu_+}\otimes\sigma_3^{\otimes\nu_3}\right)\Pi_{\rm sym}\\
&=\sum_{r_1=0}^{\nu_-}\sum_{r_2=0}^{\nu_+}\binom{\nu_-}{r_1}\binom{\nu_+}{r_2}i^{\nu_+-\nu_-+r_1-r_2}\widetilde{\sigma}_{\tilde{\boldsymbol{\nu}}(r_1+r_2)}\\
&=\sum_{r=0}^{\nu_++\nu_-}i^{\nu_+-\nu_--r}\widetilde{\sigma}_{\tilde{\boldsymbol{\nu}}(r)}\left(\sum_{q=\max(0,r-\nu_+)}^{\min(r,\nu_-)}(-1)^q\binom{\nu_-}{q}\binom{\nu_+}{r-q}\right)\\
&=\sum_{r=0}^{\nu_++\nu_-}i^{\nu_+-\nu_-+r}K_r(\nu_+,\nu_++\nu_-;2)\widetilde{\sigma}_{\tilde{\boldsymbol{\nu}}(r)}.
\end{aligned}\tag{C7}$$

with $\tilde{\boldsymbol{\nu}}(r)=(\nu_0,r,\nu_++\nu_--r,\nu_3)$. The second equality is obtained via the change of variables $r=r_1+r_2$ and $q=r_1$, and the third equality follows from the definition of the Kravchuk polynomial. Inserting this expression into (C6) yields (C2).

Appendix D: List of relevant quantities in the symmetric sector of N qubits

The averaged purity $r_t(\rho)$ is directly associated with the measures of anticoherence $\mathcal{A}_t$ based on the purity [25]

$$\mathcal{A}_t=\frac{t+1}{t}\left(1-r_t\right).\tag{D1}$$

Additionally, we define the average linear entropy over all the subsets A of t qubits

$$\mu_t(\rho)\equiv\frac{1}{\binom{N}{t}}\sum_{|A|=t}2\left[1-\mathrm{Tr}\left(\rho_A^2\right)\right].\tag{D2}$$

where ρ_A is the t-qubit density matrix obtained by tracing out the $N-t$ qubits in the complement of A. It turns out that $\mu_t(\rho)=2\left[1-r_t(\rho)\right]$.

Appendix E: Proof of some binomial identities

1. Relationship between r_t and S_k

Defining

$$C_{tk}\equiv\frac{1}{2^t}\frac{\binom{t}{k}}{\binom{N}{k}},\qquad D_{kt}\equiv\binom{N}{k}(-1)^{t+k}2^t\binom{k}{t},\tag{E1}$$

Eqs. (19) and (20) read

$$r_t=\sum_k C_{tk}S_k,\qquad S_k=\sum_t D_{kt}r_t.\tag{E2}$$

The first equation is derived, for instance, in Ref. [12] (see Eq. (31)). On the other hand, Eq. (20) follows from the identity $CD=\mathbb{1}$ which is a direct consequence of the binomial inversion theorem.

2. Relationship between r_t and a_ℓ

Let

$$Q_{t\ell}\equiv\frac{\binom{t}{\ell}\binom{N+\ell+1}{N}}{\binom{N}{\ell}\binom{t+\ell+1}{t}}\tag{E3}$$

and

$$R_{\ell t}\equiv(-1)^{t+\ell}\frac{2\ell+1}{\ell+1}\frac{\binom{N}{\ell}\binom{\ell}{t}\binom{t+\ell}{t}}{\binom{N+\ell+1}{N}}\tag{E4}$$

Equations (21) and (22) can be rewritten

$$r_t=\sum_\ell Q_{t\ell}a_\ell,\qquad a_\ell=\sum_t R_{\ell t}r_t.\tag{E5}$$

As usual, the sums run over all integers for which the summand is defined and nonzero. We want to show that $QR=\mathbb{1}$. We have

$$\sum_n Q_{in}R_{nj}=\sum_n\frac{\binom{i}{n}}{\binom{i+n+1}{i}}(-1)^{n+j}\frac{2n+1}{n+1}\binom{n}{j}\binom{n+j}{j}\tag{E6}$$

$$=\sum_n(-1)^{n+j}(2n+1)\frac{i!^2(n+j)!}{(i-n)!(i+n+1)!(n-j)!j!^2}\tag{E7}$$

Replacing $(2n+1)$ by $(i+n+1)-(i-n)$ in (E7), we get

$$\sum_n Q_{in}R_{nj}=\lambda\sum_n(-1)^{n+j}\binom{n+j}{2j}\left[\binom{2i}{i-n}-\binom{2i}{i-n-1}\right],\qquad\lambda=\frac{i!^2}{j!^2}\frac{(2j)!}{(2i)!}.\tag{E8}$$

Following [26], we define the nth Legendre polynomial P_n as

$$P_n(x)=\frac{1}{2^n}\sum_{k=0}^{\lfloor n/2\rfloor}(-1)^k\binom{n-k}{k}\binom{2n-2k}{n-k}x^{n-2k}\tag{E9}$$

(see p.66 of [26]), and we introduce the function $Q_n(x) = P_n(1+2x)$. We have ([26] Chapter 2, Problem 11, p. 78)

$$Q_n(x) = \sum_{k=0}^{n} \binom{n+k}{2k}\binom{2k}{k} x^k, \tag{E10}$$

as well as the inverse formula ([26] p. 79)

$$\sum_{k=0}^{m} (-1)^{k+m} \left[\binom{2m}{m-k} - \binom{2m}{m-k-1}\right] Q_k(x) = \binom{2m}{m} x^m. \tag{E11}$$

Summing Eq. (E8) over j and using (E10), we get

$$\sum_j \binom{2j}{j} \left(\frac{1}{\lambda}\sum_n Q_{in}R_{nj}\right) x^j = (-1)^i \sum_n (-1)^{n+i} \left[\binom{2i}{i-n} - \binom{2i}{i-n-1}\right] Q_n(-x) \tag{E12}$$

which, using (E11), yields

$$\sum_j \binom{2j}{j} \left(\frac{1}{\lambda}\sum_n Q_{in}R_{nj}\right) x^j = \binom{2i}{i} x^i \,. \tag{E13}$$

Term-by-term identification in the latter equation yields

$$\sum_n Q_{in}R_{nj} = \lambda\delta_{ij} = \delta_{ij} \,. \tag{E14}$$

3. Relationship between a_ℓ and S_k

From the above identities one gets

$$S_k = \sum_\ell \Omega_{k\ell} a_\ell, \qquad a_\ell = \sum_k \omega_{\ell k} S_k \tag{E15}$$

with $\Omega = DQ$ and $\omega = RC$. Matrix ω gives

$$\omega_{\ell k} = \sum_t R_{\ell t} C_{tk} = \frac{\binom{N}{\ell}}{\binom{N}{k}\binom{N+\ell+1}{N}} \frac{2\ell+1}{\ell+1} \sum_t \frac{(-1)^{t+\ell}}{2^t} \binom{t}{k}\binom{\ell}{t}\binom{t+\ell}{t}. \tag{E16}$$

The last sum can be obtained from the identity

$$\sum_t \frac{(-1)^{t+k}}{2^t} \binom{t}{k}\binom{\ell}{t}\binom{t+\ell}{t} = \begin{cases} 2^{-\ell}(-1)^{(\ell-k)/2} \binom{\ell+k}{(\ell+k)/2}\binom{(\ell+k)/2}{(\ell-k)/2} & \text{if } k+\ell \text{ even} \\ 0 & \text{otherwise.} \end{cases} \tag{E17}$$

Equation (E17) can be derived by expanding $Q_n((x-1)/2)$. Indeed, from (E10) we have

$$Q_\ell\left(\frac{x-1}{2}\right) = \sum_t \binom{\ell+t}{2t}\binom{2t}{t}\left(\frac{x-1}{2}\right)^t \tag{E18}$$

$$= \sum_t \binom{\ell+t}{2t}\binom{2t}{t}\left(-\frac{1}{2}\right)^t \sum_k \binom{t}{k}(-x)^k \tag{E19}$$

$$= \sum_k \left[\sum_t \frac{(-1)^{t+k}}{2^t}\binom{t}{k}\binom{\ell}{t}\binom{t+\ell}{t}\right] x^k. \tag{E20}$$

Since $Q_\ell\left(\frac{x-1}{2}\right) = P_\ell(x)$, the sum (E17) is exactly the coefficient of x^k in $P_\ell(x)$. From the definition (E9) of the Legendre polynomials, the identity (E17) follows immediately. Inserting the expression (E17) into (E16) then yields

$$\omega_{\ell k} = \frac{(-1)^{(k-\ell)/2}}{2^\ell} \frac{\binom{N}{\ell}}{\binom{N}{k}\binom{N+\ell+1}{N}} \frac{2\ell+1}{\ell+1} \binom{\ell+k}{\frac{\ell+k}{2}}\binom{\frac{\ell+k}{2}}{\frac{\ell-k}{2}} \qquad \text{for } k+\ell \text{ even, 0 otherwise.} \tag{E21}$$

Matrix Ω is given by

$$\Omega_{k\ell} = \sum_t D_{kt} Q_{t\ell} = \frac{\binom{N}{k}\binom{N+\ell+1}{N}}{\binom{N}{\ell}} \sum_t (-1)^{t+k} 2^t \binom{k}{t}\binom{t}{\ell} \frac{1}{\binom{t+\ell+1}{t}}. \tag{E22}$$

Using the identity

$$\sum_t (-1)^{t+k} 2^t \binom{k}{t}\binom{t}{\ell} \frac{1}{\binom{t+\ell+1}{t}} = \begin{cases} 2^\ell \binom{(k+\ell)/2}{(k-\ell)/2} / \binom{k+\ell+1}{k} & \text{if } k+\ell \text{ even} \\ 0 & \text{otherwise,} \end{cases} \tag{E23}$$

we get

$$\Omega_{k\ell} = 2^\ell \frac{\binom{N}{k}\binom{N+\ell+1}{N}}{\binom{N}{\ell}\binom{k+\ell+1}{k}} \binom{\frac{k+\ell}{2}}{\frac{k-\ell}{2}} \qquad \text{for } k+\ell \text{ even, 0 otherwise.} \tag{E24}$$

Identity (E23) can be proved as follows. We change variables by setting $k+\ell = 2p+\epsilon$ and $k-\ell = 2q+\epsilon$, with $\epsilon = 0$ for $k+\ell$ even and $\epsilon = 1$ for $k+\ell$ odd. Equation (E23) can be reexpressed as

$$\forall p, q, \qquad \sum_t (-1)^{t+p+q+\epsilon} 2^t \binom{p+q+\epsilon}{t}\binom{t}{p-q}\binom{t+p-q+1}{t}^{-1} = (1-\epsilon) 2^{p-q} \binom{p}{q}\binom{2p+1}{p+q}^{-1}. \tag{E25}$$

Changing the summation variable to $k = t-p+q$ and rearranging the factorials, this gives

$$\forall p, q, \qquad \sum_k (-1)^{k+\epsilon} 2^k \binom{k+p-q}{k}\binom{2p+1+\epsilon}{k-2q+2p+1} = (1-\epsilon)\binom{p}{q} \tag{E26}$$

$$\Leftrightarrow \qquad \forall p, q, \qquad \sum_k (-1)^k 2^k \binom{k+q}{k}\binom{2p+1+\epsilon}{2q+1+k} = (1-\epsilon)\binom{p}{q} \tag{E27}$$

where from line 1 to 2 we changed q to $p-q$. This equation holds trivially for $q=0$; thus it suffices to check that the left-hand side $f_{p,q}$ of (E27) verifies the Pascal triangle recursion relation. For $\epsilon = 0$ we have

$$\begin{aligned} f_{p+1,q} &= \sum_k (-1)^k 2^k \binom{k+q}{k}\binom{2p+3}{2q+1+k} \\ &= \sum_k (-1)^k 2^k \binom{k+q}{k}\left[\binom{2p+2}{2q+1+k} + \binom{2p+2}{2q+k}\right] \\ &= \sum_k (-1)^k 2^k \binom{k+q}{q}\left[\binom{2p+1}{2q+1+k} + 2\binom{2p+1}{2q+k} + \binom{2p+1}{2q+k-1}\right]. \end{aligned} \tag{E28}$$

Shifting k to $k+1$ in the third term, we get

$$\begin{aligned} \sum_k (-1)^k 2^k \binom{k+q}{q}\binom{2p+1}{2q+k-1} &= -2\sum_k (-1)^k 2^k \binom{k+1+q}{q}\binom{2p+1}{2q+k} \\ &= -2\sum_k (-1)^k 2^k \binom{k+q}{q}\binom{2p+1}{2q+k} - 2\sum_k (-1)^k 2^k \binom{k+q}{q-1}\binom{2p+1}{2q+k} \\ &= -2\sum_k (-1)^k 2^k \binom{k+q}{q}\binom{2p+1}{2q+k} + \sum_k (-1)^k 2^k \binom{k+q-1}{q-1}\binom{2p+1}{2q+k-1}, \end{aligned} \tag{E29}$$

where in the last term we have changed back k to $k-1$. Thus Eq. (E28) becomes

$$f_{p+1,q} = \sum_k (-1)^k 2^k \binom{k+q}{q}\binom{2p+1}{2q+1+k} + \sum_k (-1)^k 2^k \binom{k+q-1}{q-1}\binom{2p+1}{2q+k-1} = f_{p,q} + f_{p,q-1}, \tag{E30}$$

which yields the desired relation. The case $\epsilon = 1$ is treated in the same way, which proves (E23).

4. Connection with special polynomials

From the discussion in the previous subsection, we have

$$\omega_{\ell k} = \frac{\binom{N}{\ell}}{\binom{N}{k}\binom{N+\ell+1}{N}} \frac{2\ell+1}{\ell+1} \times \text{Coeff. of } x^k \text{ in } P_\ell(x). \tag{E31}$$

In other words, ω can be defined by the generating function

$$\sum_k \binom{N}{k} \omega_{\ell k}\, x^k = \frac{\binom{2N}{N+\ell} - \binom{2N}{N+\ell+1}}{\binom{2N}{N}} P_\ell(x). \tag{E32}$$

One can also obtain an analog of Eq. (E32) for Ω by using the Chebyshev polynomials of the second kind defined by

$$U_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \binom{n-k}{k} (2x)^{n-2k}. \tag{E33}$$

Indeed, replacing the above sum over k by a sum over $\ell = n-2k$ and making the substitution $n \to k$ we get

$$U_k(x) = \sum_{\ell; k+\ell \text{ even}} (-1)^{\frac{k-\ell}{2}} 2^\ell \binom{\frac{k+\ell}{2}}{\frac{k-\ell}{2}} x^\ell. \tag{E34}$$

Using this form (E34), the expression (E24) for $\Omega_{k\ell}$ gives

$$\sum_\ell (-1)^{\frac{k-\ell}{2}} \frac{\binom{k+\ell+1}{k}}{\binom{N+\ell+1}{N}} \binom{N}{\ell} \Omega_{k\ell}\, x^\ell = \binom{N}{k} U_k(x). \tag{E35}$$

Note that if we set $\tilde{\Omega}_{kp} = \Omega_{k,k-2p}$ and $\tilde{\omega}_{\ell q} = (-1)^q \omega_{\ell,\ell-2q}$, Eq. (E15) can be rewritten

$$S_k = \sum_p \tilde{\Omega}_{kp} a_{k-2p}, \qquad a_\ell = \sum_q (-1)^q \tilde{\omega}_{\ell q} S_{\ell-2q}, \tag{E36}$$

which is exactly the form of a Chebyshev inverse pair (see last equation at p.67 of [26]).

Appendix F: Relation among $\mathcal{T}_L$ and $\mathcal{P}_j$

The relation among the operators $\mathcal{T}_L$ and $\mathcal{P}_j$ is (Appendix C of Ref. [13])

$$\mathcal{T}_L = (2L+1) \sum_{j=0}^{N} (-1)^{N+j} \begin{Bmatrix} N/2 & N/2 & j \\ N/2 & N/2 & L \end{Bmatrix} \mathcal{P}_j, \qquad \mathcal{P}_j = (-1)^{N+j} (2j+1) \sum_{L=0}^{N} \begin{Bmatrix} N/2 & N/2 & L \\ N/2 & N/2 & j \end{Bmatrix} \mathcal{T}_L. \tag{F1}$$

From these expressions, we can extract the expressions of the p_j's in terms of a_L's (39), and viceversa, by evaluating in the double-copy state $|\Psi\rangle = |\psi\rangle \otimes |\psi\rangle$.

Appendix G: Examples for small spin

Case $j = 5/2$

$$\begin{aligned}
a_1 &= \frac{1}{70}\left(-31p_1 - 11p_3 + 25p_5\right), \\
a_2 &= \frac{1}{84}\left(46p_1 - 29p_3 + 25p_5\right), \\
a_3 &= \frac{1}{180}\left(-66p_1 + 79p_3 + 25p_5\right), \\
a_4 &= \frac{1}{28}\left(-6p_1 + 19p_3 + p_5\right), \\
a_5 &= \frac{1}{252}\left(330p_1 + 55p_3 + p_5\right).
\end{aligned} \tag{G1}$$

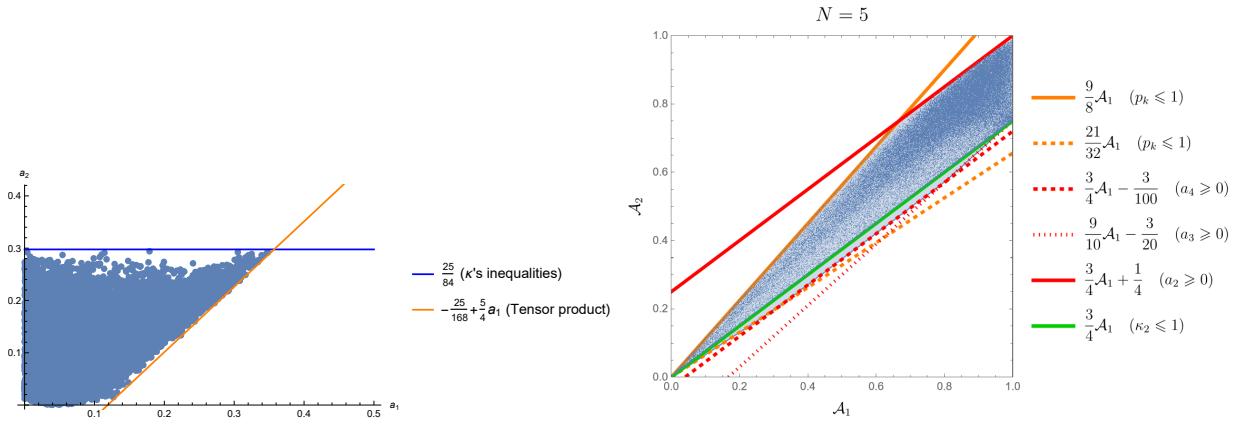


FIG. 3. Numerical range of a_2 vs a_1 for $N = 5$ ($j = 5/2$). The orange line is given by the analysis in the $\hat{\mathcal{Y}}_L$, while the blue boundary is given by the κ_1, $\kappa_2 \leqslant 1$, which are the inequalities where we already have found a proof.

Again, there is a linear combination between a_2 and a_1 that gives the right inequality

$$4a_2 - 5a_1 = -\frac{25}{42} + 5p_1 \geqslant -\frac{25}{42}\,. \tag{G2}$$

The purities of the reduced states and the anticoherence measures read

$$\begin{aligned} r_1 =& 1 - \frac{28}{25}p_1 - \frac{18}{25}p_3\,, & \mathcal{A}_1 =& \frac{4}{25}\left(14p_1 + 9p_3\right)\,, \\ r_2 =& 1 - \frac{49}{50}p_1 - \frac{27}{25}p_3\,, & \mathcal{A}_2 =& \frac{3}{100}\left(49p_1 + 54p_3\right)\,. \end{aligned} \tag{G3}$$

In particular,

$$\frac{25}{18}r_1 - \frac{25}{27}r_2 = \frac{25}{54} - \frac{35}{54}p_1 \leqslant \frac{25}{54}\,, \quad \Longrightarrow \quad 6r_1 - 4r_2 \leqslant 2\,, \tag{G4}$$

and

$$9\mathcal{A}_1 - 8\mathcal{A}_2 = \frac{42}{5}p_1 \geqslant 0\,, \tag{G5}$$

which gives the orange solid line in Fig. 3. Again, this proves the r.h.s. of the inequality (I2).

Case $j = 3$

$$\begin{aligned} a_1 =& \frac{1}{28}\left(-12p_0 - 9p_2 - 2p_4 + 9p_6\right)\,, \\ a_2 =& \frac{1}{84}\left(60p_0 + 19p_2 - 30p_4 + 25p_6\right)\,, \\ a_3 =& \frac{1}{6}\left(-6p_0 + p_2 + p_4 + p_6\right)\,, \\ a_4 =& \frac{1}{154}\left(198p_0 - 99p_2 + 97p_4 + 9p_6\right)\,, \\ a_5 =& \frac{1}{84}\left(-132p_0 + 55p_2 + 34p_4 + p_6\right)\,, \\ a_6 =& \frac{1}{924}\left(1716p_0 + 715p_2 + 78p_4 + p_6\right)\,. \end{aligned} \tag{G6}$$

In particular,

$$\mathcal{A}_1 = \frac{7}{3}p_0 + 2p_2 + \frac{11}{9}p_4\,, \quad \mathcal{A}_2 = \frac{7}{5}p_0 + \frac{39}{25}p_2 + \frac{22}{15}p_4\,, \tag{G7}$$

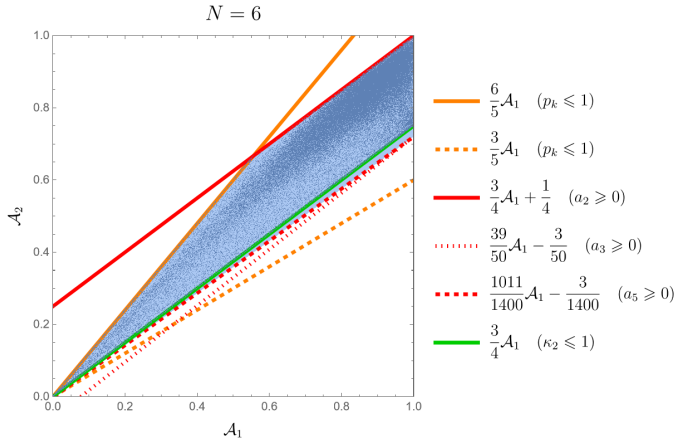


FIG. 4. Numerical range of a_2 vs a_1 for $N = 6$ ($j = 3$).

and

$$6\mathcal{A}_1 - 5\mathcal{A}_2 \geqslant 0\,. \tag{G8}$$

Again, this proves the r.h.s. of the inequality (I2).

Appendix H: Attempts to prove the general case

1. Proof for the general case

We want to proof the l.h.s.of Eq. (I3),

$$(N-t)tr_{t+1} + N - 2t - 1 - (N-t-1)(t+1)r_t \geqslant 0\,, \tag{H1}$$

using the previous results. A direct calculation shows us that

$$r_t = (N+1)^2 \sum_{\substack{\alpha=0 \\ \alpha\equiv N(\mathrm{mod}2)}}^{N} p_\alpha \begin{Bmatrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2} \\ \frac{N}{2} & \alpha & \frac{N}{2} \\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2} \end{Bmatrix} = \frac{t!(N-t)!}{N!^2} \sum_{\substack{\alpha=0 \\ \alpha\equiv N(\mathrm{mod}2)}}^{N} \sqrt{\frac{(N+\alpha+1)!(N-\alpha)!}{2\alpha+1}} C^{\alpha 0}_{j\,t-j,j\,j-t} p_\alpha\,, \tag{H2}$$

where the 9j-symbols appear. In particular, for $\alpha = N$ and $N-2$, the 9j-symbol are

$$\begin{Bmatrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2} \\ \frac{N}{2} & \frac{N}{2} & \frac{N}{2} \\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2} \end{Bmatrix} = \frac{1}{(N+1)^2}\,, \qquad \begin{Bmatrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2} \\ \frac{N}{2} & N-2 & \frac{N}{2} \\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2} \end{Bmatrix} = \frac{N^3 - N^2(4t+1) + 2Nt(2t+1) - 2t^2}{(N-1)N^2(N+1)^2}\,, \tag{H3}$$

After we substitute

$$p_N = 1 - \sum_{\substack{\alpha=0 \\ \alpha\equiv N(\mathrm{mod}2)}}^{N-2} p_\alpha \tag{H4}$$

in r_t, we get

$$r_t = 1 + \sum_{\substack{\alpha=0 \\ \alpha\equiv N(\mathrm{mod}2)}}^{N-2} p_\alpha \left((N+1)^2 \begin{Bmatrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2} \\ \frac{N}{2} & \alpha & \frac{N}{2} \\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2} \end{Bmatrix} - 1 \right) \equiv 1 + \sum_{\substack{\alpha=0 \\ \alpha\equiv N(\mathrm{mod}2)}}^{N-2} p_\alpha K(N,t,\alpha)\,. \tag{H5}$$

For, $\alpha = N-2$

$$K(N,t,N-2) = -\frac{2(2N-1)t(N-t)}{(N-1)N^2}\,. \tag{H6}$$

In particular, we have observed numerically that the following inequality is fulfilled

$$\begin{aligned}
&K(N,t+1,N-2)r_t - K(N,t,N-2)r_{t+1} = \frac{2(2N-1)}{(N-1)N^2}\Big[t(N-t)r_{t+1} - (t+1)(N-t-1)r_t\Big] \\
&= -\frac{2(2N-1)}{(N-1)N^2}(N-2t-1) + \sum_{\substack{\alpha=0 \\ \alpha\equiv N(\mathrm{mod}2)}}^{N-4} p_\alpha\Big[K(N,t+1,N-2)K(N,t,\alpha) - K(N,t,N-2)K(N,t+1,\alpha)\Big] \qquad \text{(H7)} \\
&\geqslant -\frac{2(2N-1)}{(N-1)N^2}(N-2t-1)\,.
\end{aligned}$$

With $j = N/2$, the following expressions for D are exact; their nonnegativity for all allowed parameters is conjectured:

$$\begin{aligned}
D =& \frac{(N-1)N^2}{2(2N-1)}\Big[K(N,t+1,N-2)K(N,t,\alpha) - K(N,t,N-2)K(N,t+1,\alpha)\Big] \\
=& N-2t-1+(N+1)^2\left[t(N-t)\begin{Bmatrix} \frac{t+1}{2} & \frac{N}{2} & \frac{N-t-1}{2} \\ \frac{N}{2} & \alpha & \frac{N}{2} \\ \frac{N-t-1}{2} & \frac{N}{2} & \frac{t+1}{2} \end{Bmatrix} - (t+1)(N-t-1)\begin{Bmatrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2} \\ \frac{N}{2} & \alpha & \frac{N}{2} \\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2} \end{Bmatrix}\right] \qquad \text{(H8)} \\
=& N-2t-1+\frac{(t+1)!(N-t)!}{N!^2}\sqrt{\frac{(N+\alpha+1)!(N-\alpha)!}{2\alpha+1}}\Big[tC^{\alpha 0}_{j\,t-j+1,j\,j-t-1} - (N-t-1)C^{\alpha 0}_{j\,t-j,j\,j-t}\Big] \\
\geqslant& 0
\end{aligned}$$

for all $N \geqslant 2$, $0 \leqslant \alpha \leqslant N$ with α and N of the same parity, and all t, $1 \leqslant t \leqslant \frac{N-1}{2}$.

In fact one has

$$\tilde{D} \equiv \binom{N}{t}\binom{N}{t+1}D = t(t+1)Z_{t+1} - (N-t)(N-t-1)Z_t + \binom{N}{t}\binom{N}{t+1}(N-2t-1), \qquad \text{(H9)}$$

with

$$Z_t = \sum_{k=\max(t+\alpha-N,0)}^{\min(\alpha,t)} (-1)^{t+k}\binom{N-\alpha}{t-k}\binom{\alpha}{k}^2. \qquad \text{(H10)}$$

Changing k to $t-k$ in the summation gives

$$Z_t = \sum_{k=\max(t-\alpha,0)}^{\min(N-\alpha,t)} (-1)^{k}\binom{N-\alpha}{k}\binom{\alpha}{t-k}^2. \qquad \text{(H11)}$$

Set $q = (N-\alpha)/2 \in \{0,\ldots,\lfloor N/2\rfloor\}$, so that the total spin remains $\alpha = N-2q$. The proposed inequality $\tilde{D} \geqslant 0$ can be studied using

$$Z_t = \sum_{k=\max(t+2q-N,0)}^{\min(2q,t)} (-1)^{k}\binom{2q}{k}\binom{N-2q}{t-k}^2 \qquad \text{(H12)}$$

$$= \sum_{k=\max(0,2q-t)}^{\min(N-t,2q)} (-1)^{k}\binom{2q}{k}\binom{N-2q}{t-2q+k}^2 \qquad \text{(H13)}$$

where in the last sum we changed k to $2q-k$. For $0 \leqslant t \leqslant N-2q$, Eq. (H13) can be rewritten

$$Z_t = \binom{N-2q}{t}^2 {}_3F_2\left(\begin{matrix}-2q,-t,-t\\ c,c\end{matrix};1\right), \qquad c = N-2q-t+1. \qquad \text{(H14)}$$

For larger t, the finite binomial sum remains the defining expression; a product of a vanishing binomial factor and a singular hypergeometric function is not evaluated directly. Using the terminating identity [27]

$${}_3F_2\left(\begin{matrix}-n,b,c\\ d,e\end{matrix};1\right) = \frac{(-1)^n(e-b)_n(e-c)_n}{(d)_n(e)_n}{}_3F_2\left(\begin{matrix}-n,1-e-n,b+c-d-e-n+1\\ b-e-n+1,c-e-n+1\end{matrix};1\right) \qquad \text{(H15)}$$

valid for nonnegative integer n and nonsingular denominator parameters (with singular cases treated by limits), Z_t simplifies to

$$Z_t = \binom{N}{t}^2 {}_3F_2\left(\begin{matrix}-2q, -N+t, 2q-2N-1\\ -N, -N\end{matrix};1\right), \tag{H16}$$

where the ${}_3F_2$ function is a polynomial of degree $2q$ in t. Using [28]

$${}_3F_2\left(\begin{matrix}-n, b, c\\ d, e\end{matrix};1\right) = \frac{(d-b)_n}{(d)_n} {}_3F_2\left(\begin{matrix}-n, b, e-c\\ e, b-d-n+1\end{matrix};1\right) \tag{H17}$$

gives

$$Z_t = \binom{N}{t}^2 {}_3F_2\left(\begin{matrix}-2q, -t, 2q-2N-1\\ -N, -N\end{matrix};1\right). \tag{H18}$$

Hence, Eq. H9 yields

$$\tilde{D} = t(t+1)\binom{N}{t+1}^2 {}_3F_2\left(\begin{matrix}-2q, -t-1, 2q-2N-1\\ -N, -N\end{matrix};1\right) \tag{H19}$$

$$-(N-t)(N-t-1)\binom{N}{t}^2 {}_3F_2\left(\begin{matrix}-2q, -N+t, 2q-2N-1\\ -N, -N\end{matrix};1\right) + \binom{N}{t}\binom{N}{t+1}(N-2t-1), \tag{H20}$$

or equivalently

$$D = t(N-t)P(t+1) - (t+1)(N-t-1)P(t), \qquad P(t) = {}_3F_2\left(\begin{matrix}-2q, -t, 2q-2N-1\\ -N, -N\end{matrix};1\right) - 1 \tag{H21}$$

where $P(t)$ is a polynomial in t. This polynomial $P(t)$ is divisible by t, and hence, by symmetry, by $N-t$. Therefore D is divisible (as a polynomial in t) by $t(N-t)$. From the same reasoning for $P(t+1)$, D is also divisible by $(t+1)(N-t-1)$. Let $R(t)$ be the polynomial $R(t) = \frac{1}{t(N-t)}P(t)$. We have

$$\frac{D}{t(N-t)(t+1)(N-t-1)} = R(t+1) - R(t). \tag{H22}$$

Showing $D \geqslant 0$ thus amounts to showing that $R(t+1) \geqslant R(t)$ for $1 \leqslant t \leqslant \lfloor (N-1)/2 \rfloor$.

The finite binomial expression for the recoupling coefficient gives

$$(N+1)^2 \begin{Bmatrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2} \\ \frac{N}{2} & N-2q & \frac{N}{2} \\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2} \end{Bmatrix} = {}_3F_2\left(\begin{matrix}-2q, -t, 2q-2N-1\\ -N, -N\end{matrix};1\right) \equiv Q_{2q}(t; -M, -M, N) \tag{H23}$$

with $M = N+1$, which turns out to be a Hahn polynomial (Eq. 1.5.1 of [29]) (in fact, since $M > N$, these are called Hahn-Eberlein polynomials). The "difference equation" 1.5.4 of [29] relates $Q_{2q}(t+1)$, $Q_{2q}(t)$ and $Q_{2q}(t-1)$; it reads

$$2q(2q-2N-1)Q_{2q}(t) = (N-t)^2 Q_{2q}(t+1) - [t^2 + (N-t)^2]Q_{2q}(t) + t^2 Q_{2q}(t-1). \tag{H24}$$

In terms of $Q_{2q}(t) \equiv Q_{2q}(t; -M, -M, N)$ we have

$$D = N - 2t - 1 + t(N-t)Q_{2q}(t+1) - (t+1)(N-t-1)Q_{2q}(t). \tag{H25}$$

From (H22), showing $D \geqslant 0$ is equivalent to showing that $R(t) = \frac{Q_{2q}(t)-1}{t(N-t)}$ increases with t.

2. Another expansion

Here α again denotes total spin, with $\alpha \equiv N \pmod 2$, and $j = N/2$.

$$\begin{aligned} D =& N-2t-1+\frac{(t+1)!(N-t)!}{N!^2}\sqrt{\frac{(N+\alpha+1)!(N-\alpha)!}{2\alpha+1}}\left[tC^{\alpha 0}_{j\,t-j+1,j\,j-t-1}-(N-t-1)C^{\alpha 0}_{j\,t-j,j\,j-t}\right] \\ =& N-2t-1+\frac{(-1)^{t+1}(t+1)!(N-t)!}{N!}\times \\ & \left[t\binom{t+\alpha+1}{\alpha}\,{}_3F_2\left(\begin{matrix}-t-1,-N-\alpha-1,-\alpha\\ -N,-t-\alpha-1\end{matrix};1\right)+(N-t-1)\binom{t+\alpha}{\alpha}\,{}_3F_2\left(\begin{matrix}-t,-N-\alpha-1,-\alpha\\ -N,-t-\alpha\end{matrix};1\right)\right] \\ \geqslant& 0 \end{aligned} \tag{H26}$$

which we conjecture that it is fulfilled for any $N = 2j > 3$, $0 < t < \frac{N}{2} - 1$ and $0 \leqslant \alpha \leqslant N-4$ with $\alpha \equiv N(\mod 2)$.

$$D = N-2t-1+(-1)^{t+1}\left[\frac{t(N-t)}{\binom{N}{t+1}^2}R_{t+1}+\frac{(t+1)(N-t-1)}{\binom{N}{t}^2}R_t\right] \geqslant 0\,. \tag{H27}$$

with

$$R_t = \sum_{k=0}^{\min(\alpha,t)}(-1)^k\binom{N-\alpha}{t-k}\binom{\alpha}{k}^2. \tag{H28}$$

3. P_α in terms of T_{LM}

$$\mathcal{P}_\tau = (-1)^{2j+\tau}(2\tau+1)\sum_{L,M}\left\{\begin{matrix} j & j & L\\ j & j & \tau\end{matrix}\right\}T_{LM}\otimes T^\dagger_{LM}\,. \tag{H29}$$

4. Attempt to proof $D \geqslant 0$ via operators

$$\begin{aligned} D =& N-2t-1+\frac{(t+1)!(N-t)!}{N!^2}\sqrt{\frac{(N+\alpha+1)!(N-\alpha)!}{2\alpha+1}}\left[tC^{\alpha 0}_{j\,t-j+1,j\,j-t-1}-(N-t-1)C^{\alpha 0}_{j\,t-j,j\,j-t}\right] \\ =& N-2t-1+\frac{(t+1)!(N-t)!}{N!C^{\alpha 0}_{j-j,jj}}\left[\frac{(N-t)!}{N!(t+1)!}\right]\langle j,j,J=\alpha,M=0|\left(\frac{t}{N-t}(J_{1+}J_{2-})-(N-t-1)(t+1)\right)(J_{1+}J_{2-})^t|j,-j\rangle|jj\rangle \end{aligned} \tag{H30}$$

For arbitrary nonorthogonal vectors, the valid bound is

$$\left|\frac{v_1^\dagger A v_2}{v_1^\dagger v_2}\right| \leqslant \frac{\|A\|_{\mathrm{op}}\,\|v_1\|\,\|v_2\|}{|v_1^\dagger v_2|}. \tag{H31}$$

There is no bound by the largest eigenvalue alone for distinct vectors: for $A = \mathrm{diag}(1,0)$, $v_1 = (1,1)^T$, and $v_2 = (1,-1+\varepsilon)^T$, the quotient is $1/\varepsilon$. Thus this proposed step does not establish $D \geqslant 0$.

$$\boldsymbol{J}^2 - \boldsymbol{J}_1^2 - \boldsymbol{J}_2^2 - 2J_{1z}J_{2z} = J_{1+}J_{2-}+J_{1-}J_{2+}\,, \tag{H32}$$

$$J_{1-}J_{2+}\,|jm\rangle\,|j-m\rangle = (j+m)(j-m+1)\,|j,m-1\rangle\,|j,-m+1\rangle \tag{H33}$$

5. 9j-symbols

$$\left\{\begin{matrix} \frac{t}{2} & \frac{N}{2} & \frac{N-t}{2}\\ \frac{N}{2} & \alpha & \frac{N}{2}\\ \frac{N-t}{2} & \frac{N}{2} & \frac{t}{2}\end{matrix}\right\} = \sum_{x=0}^{t}F(x,t,N)\left\{\begin{matrix} \frac{N}{2} & \frac{N}{2} & \alpha\\ \frac{N}{2} & \frac{N}{2} & x\end{matrix}\right\} \tag{H34}$$

with

$$F(x,t,N)=(2x+1)\left\{\begin{array}{ccc}\frac{N-t}{2} & \frac{t}{2} & \frac{N}{2}\\ x & \frac{N}{2} & \frac{t}{2}\end{array}\right\}^2=(2x+1)\frac{t!^2(N-x)!(N+x+1)!}{(N+1)!^2(t-x)!(t+1+x)!} \tag{H35}$$

The function $F(x,t,N)$ is always positive and fulfills that

$$\sum_{x=0}^{t}F(x,t,N)=\frac{1}{N-t+1}\,. \tag{H36}$$

Then

$$D=N-2t-1+(N+1)^2\left[t(N-t)\sum_{x=0}^{t+1}F(x,t+1,N)\left\{\begin{array}{ccc}\frac{N}{2} & \frac{N}{2} & \alpha\\ \frac{N}{2} & \frac{N}{2} & x\end{array}\right\}-(t+1)(N-t-1)\sum_{x=0}^{t}F(x,t,N)\left\{\begin{array}{ccc}\frac{N}{2} & \frac{N}{2} & \alpha\\ \frac{N}{2} & \frac{N}{2} & x\end{array}\right\}\right] \tag{H37}$$

Appendix I: Conjectures for the bounds of anticoherent measures

Observation 1: For pure symmetric states with $N=2j$ and integer $1\leqslant t\leqslant\lfloor(N-1)/2\rfloor$, Conjecture (3) of [30] is given by the inequalities

$$\frac{t(t+2)}{(t+1)^2}\mathcal{A}_t\leqslant\mathcal{A}_{t+1}\leqslant\frac{(2j-t-1)(t+2)}{(2j-t)(t+1)}\mathcal{A}_t\,, \tag{I1}$$

can be rewritten as

$$\frac{t}{(t+1)}\mathcal{A}_t\leqslant\frac{(t+1)}{(t+2)}\mathcal{A}_{t+1}\leqslant\frac{(2j-t-1)}{(2j-t)}\mathcal{A}_t\,. \tag{I2}$$

These inequalities in terms of the purity of the state $r_t=\mathrm{Tr}(\rho_t^2)$ read

$$\frac{2t+1-2j+(2j-t-1)(t+1)r_t}{(2j-t)t}\leqslant r_{t+1}\leqslant r_t\,, \tag{I3}$$

where the r.h.s. is equal to the l.h.s. of Eq. (I1) and viceversa.

Observation 2: For $1\leqslant t<t\prime\leqslant\lfloor(N+1)/2\rfloor$, if Eq. (I1) is true, then Eq. (5) of [30] is also true. In fact, it is a sequence of inequalities

$$\left(\frac{t}{t+1}\right)\mathcal{A}_t\leqslant\left(\frac{t+1}{t+2}\right)\mathcal{A}_{t+1}\leqslant\cdots\leqslant\left(\frac{t'}{t'+1}\right)\mathcal{A}_{t'}\,, \tag{I4}$$

$$\frac{\mathcal{A}_t}{(2j-t)(t+1)}\geqslant\frac{\mathcal{A}_{t+1}}{(2j-t-1)(t+2)}\geqslant\frac{\mathcal{A}_{t+2}}{(2j-t-2)(t+3)}\geqslant\cdots\geqslant\frac{\mathcal{A}_{t'}}{(2j-t')(t'+1)}\,. \tag{I5}$$

Observation 3: It is mentioned as conjecture in Ref. [30] that

$$\begin{aligned}S_1&\leqslant\binom{N}{1}\,,\\ S_2&\leqslant\binom{N}{2}\,,\\ &\vdots\end{aligned} \tag{I6}$$

at least when fewer than half of the indices are nonzero. The bounds for S_1 and S_2 are proved in Ref. [12]. For each fixed sector S_m, Theorem 2 of that reference establishes an eventual threshold $N_0'(m)$ beyond which $S_m\leqslant\binom{N}{m}$; it does not assert this bound for the full sector S_N. For $m=3$, $N_0'=5$ [6]; for $m=4$, $7\leq N_0'(4)\leq 9$, as summarized in Proposition 1.

For the highest sectors, the statements must be distinguished by parity:

$$\begin{aligned} S_N &\geqslant 1 \quad \text{(proved)}, \\ S_{N-1} &\geqslant N \quad (N \text{ odd; conjectured in general}), \\ \min_{\psi\in\mathcal{H}^{\vee N}} S_{N-1} &= 0 \quad (N \text{ even}). \end{aligned} \tag{I7}$$

The first bound is saturated by a coherent product state, and the latter two by $|\mathrm{GHZ}(N)\rangle$; the odd-N bound is proved here for $N = 9$. A proposed bound $S_{N-2} \geqslant N-1$ for even $N \geqslant 4$ concerns a different sector and is not a continuation of a universal binomial lower bound. A product $|\mathrm{GHZ}(N-1)\rangle \otimes |0\rangle$ attains $N-1$ in the unrestricted space but is not fully symmetric and cannot establish attainability in the symmetric sector.

Appendix J: Nonlinear inequalities

1. Quartic multipole moments and a quadratic constraint on the r_t

We start with the expansion of the symmetric state in the irreducible tensor operator basis

$$\rho = \sum_{L=0}^{N} \sum_{M=-L}^{L} \rho_{LM}\, T_{LM}, \qquad \rho_{LM} = \mathrm{Tr}(\rho\, T_{LM}^{\dagger}), \tag{J1}$$

and define the quadratic moments

$$a_L = \sum_{M=-L}^{L} |\rho_{LM}|^2 \geq 0. \tag{J2}$$

We now introduce the quartic (fourth-power) moments

$$b_L = \sum_{M=-L}^{L} |\rho_{LM}|^4 \geq 0, \qquad B = \sum_{L=0}^{N} b_L = \sum_{L,M} |\rho_{LM}|^4. \tag{J3}$$

By Cauchy–Schwarz in each L-sector, for $x_M = |\rho_{LM}|^2$,

$$\Big(\sum_M 1\, x_M\Big)^2 \leq (2L+1) \sum_M x_M^2 \quad \Rightarrow \quad b_L \geq \frac{a_L^2}{2L+1}. \tag{J4}$$

Moreover, using $B = \sum_i x_i^2 \leq (\sum_i x_i)^2$ for $x_i \geq 0$ and $\sum_{L,M} |\rho_{LM}|^2 = \mathrm{Tr}(\rho^2)$, we obtain for pure states $\mathrm{Tr}(\rho^2) = 1$:

$$\sum_{L=0}^{N} \frac{a_L^2}{2L+1} \leq B \leq \Big(\sum_{L,M} |\rho_{LM}|^2\Big)^2 = \mathrm{Tr}(\rho^2)^2 = 1, \qquad \Rightarrow \qquad \sum_{L=0}^{N} \frac{a_L^2}{2L+1} \leq 1. \tag{J5}$$

Finally, using the inverse linear relation between a_L and the reduced purities r_t,

$$a_L = \sum_{t=0}^{L} R_{Lt}\, r_t, \tag{J6}$$

we obtain an explicit quadratic inequality in the r_t for pure symmetric states:

$$\boxed{\sum_{L=0}^{N} \frac{1}{2L+1} \left(\sum_{t=0}^{L} R_{Lt}\, r_t\right)^2 \leq 1.} \tag{J7}$$

Appendix K: Nonlinear inequalities for pure states

The pure-state condition is $\rho^2 = \rho$. For $N = 5$ and $i = 1, 2, 3$, expanding $\mathrm{Tr}(\rho^2 \sigma_{i\mathbf{0}_4}) = \mathrm{Tr}(\rho \sigma_{i\mathbf{0}_4})$ in the Pauli

basis gives the exact identity

$$16x_{i\mathbf{0}_4} = \sum_{\nu_1,\ldots,\nu_4=0}^{3} x_{i\nu_1\nu_2\nu_3\nu_4}\, x_{0\nu_1\nu_2\nu_3\nu_4}. \tag{K1}$$

For a symmetric state, define $\boldsymbol{\eta}(\boldsymbol{\nu}) = (\mathbf{0}_{\nu_0}\mathbf{1}_{\nu_1}\mathbf{2}_{\nu_2}\mathbf{3}_{\nu_3})$. Grouping the four remaining indices by their multiplicities yields

$$16x_{i\mathbf{0}_4} = \sum_{\substack{\nu_0,\nu_1,\nu_2,\nu_3\geqslant 0\\ \nu_0+\nu_1+\nu_2+\nu_3=4}} \frac{4!}{\nu_0!\nu_1!\nu_2!\nu_3!}\, x_{i\boldsymbol{\eta}(\boldsymbol{\nu})} x_{0\boldsymbol{\eta}(\boldsymbol{\nu})}. \tag{K2}$$

This sum counts each ordering once and includes the multiplicity pattern $(2,2)$.

Now, for Hermitian A, B, we use the Cauchy-Schwarz inequality of the quantum covariance

$$|\mathrm{Cov}_\rho(A,B)|^2 \leqslant \mathrm{Var}_\rho(A)\mathrm{Var}_\rho(B) \tag{K3}$$

$$|\mathrm{Tr}(\rho AB) - \mathrm{Tr}(\rho A)\mathrm{Tr}(\rho B)|^2 \leqslant \left(\mathrm{Tr}(\rho A^2) - (\mathrm{Tr}\rho A)^2\right) \times \left(\mathrm{Tr}(\rho B^2) - (\mathrm{Tr}\rho B)^2\right), \tag{K4}$$

Appendix L: Stationarity and C_2 symmetry for five-qubit extrema

Using the expression (9) for $x_{\boldsymbol{\mu}}$, we can rewrite the sector lengths for symmetric pure states (11) as

$$\begin{aligned} S_m &= \sum_{\mathrm{wt}(\sigma_{\boldsymbol{\mu}})=m} \langle\psi|\, L_{\boldsymbol{\mu}}\, |\psi\rangle^2 \\ &= \binom{N}{m} \sum_{a_1,\ldots,a_m=1}^{3} \langle\psi|\, L_{a_1\ldots a_m 0\ldots 0}\, |\psi\rangle^2, \end{aligned} \tag{L1}$$

In particular, we have

$$S_1 = N\sum_{a=1}^{3} v_a^2, \quad v_a \equiv \langle\psi|\, L_a\, |\psi\rangle \tag{L2}$$

and

$$S_2 = \binom{N}{2} \mathrm{tr}\, Q^2, \quad Q_{ab} \equiv \langle\psi|\, L_{ab}\, |\psi\rangle \tag{L3}$$

Here $L_a \equiv L_{a0\ldots0}$ and $L_{ab} \equiv L_{ab0\ldots0}$. From [1] we can express L_a and L_{ab} in terms of $(N+1)$-dimensional angular momentum matrices J_x, J_y and J_z, namely

$$L_a = \frac{2}{5} J_a \tag{L4}$$

and

$$L_{ab} = \frac{1}{10}\left(J_aJ_b + J_bJ_a\right) - \frac{\delta_{ab}}{4}\mathbf{1}. \tag{L5}$$

Matrix Q is symmetric since $L_{\boldsymbol{\mu}}$ are invariant under permutation of indices.

The Majorana representation is covariant under collective unitary transformations: applying $U^{\otimes N}$ rotates all points rigidly. Sector lengths are invariant under these rotations, so an extremal symmetric state has an entire collective-unitary orbit of extremizers within the symmetric sector.

Crucially, the basis matrices $L_{\boldsymbol{\mu}}$ are covariant, in the sense that for any transformation Λ of the Lorentz group, we have

$$\mathcal{D}(\Lambda)\, L_{\mu_1\mu_2\ldots\mu_N}\, \mathcal{D}(\Lambda)^\dagger = \Lambda^{\nu_1}_{\mu_1}\ldots\Lambda^{\nu_N}_{\mu_N}\, L_{\nu_1\nu_2\ldots\nu_N}. \tag{L6}$$

with $\mathcal{D}(\Lambda)$ the $(N+1)$-dimensional matrix representing Λ in the $(N/2,0)$ representation [1]. As a consequence, the vector and matrix appearing in (L2) and (L3) transform as tensors under rotations. Namely, if $\Lambda \in SO(3)$ rotates the Majorana points of a state $|\psi\rangle$ and brings it to a state $|\psi^\Lambda\rangle$, then $v(\psi^\Lambda) = \Lambda v(\psi)$ and $Q(\psi^\Lambda) = \Lambda Q(\psi)\Lambda^T$. Since $S_1 \propto ||v||^2$ and $S_2 \propto \mathrm{tr}\, Q^2$ are invariant under such rotations, for any extremal state $|\psi\rangle$ there exists a rotation Λ that transforms $|\psi\rangle$ into an extremal state $|\psi^\Lambda\rangle$ such that the vector $v(\psi)$ is brought to a vector $v(\psi^\Lambda)$ parallel to the z axis. We can therefore, without loss of generality, search for extremal states such that $v_x = v_y = 0$.

Let us introduce the operators

$$W = \sum_a v_a L_a, \tag{L7}$$

$$H = \sum_{a,b} Q_{ab} L_{ab} \tag{L8}$$

and the vectors

$$X_c = 2\,\langle\psi|\, L_c H + H L_c\, |\psi\rangle \tag{L9}$$

$$Y_c = \langle\tilde{\psi}|\, L_c\, |\psi\rangle\,, \tag{L10}$$

with $|\psi\rangle$ some symmetric state and $|\tilde{\psi}\rangle$ its time-reversed (18).

By using the explicit expressions of the angular momentum 6-dimensional representation matrices J_x, J_y, J_z, we can calculate explicity all quantities v_a, Q_{ab}, the 6×6 matrices W and H as well as the 3-dimensional vectors X and Y, for an arbitrary symmetric 5-qubit state written in the Dicke basis $|\psi\rangle = \sum_{k=0}^{5} z_k\, |D_5^{(k)}\rangle$, as a polymonial in the complex variables z_k. It can easily be checked by explicit calculation that the following identity holds for all z_k:

$$X = \frac{16}{5} Q v + \frac{4}{5}\langle\psi|\psi\rangle\, v + \frac{4i}{5} Y \times Y^*. \tag{L11}$$

Explicitly, we get $Y = (i(f-g),\, f+g,\, -ih)$ with

$$\begin{aligned} f &= \frac{1}{5}\left(3z_3^2 - 4\sqrt{2}z_2z_4 + 2\sqrt{5}z_1z_5\right) \\ g &= \frac{1}{5}\left(3z_2^2 - 4\sqrt{2}z_1z_3 + 2\sqrt{5}z_0z_4\right) \\ h &= -\frac{2}{5}z_2z_3 + \frac{6z_1z_4}{5} - 2z_0z_5. \end{aligned} \tag{L12}$$

The tensor product of two spins-5/2 decomposes over spins 0 to 5. The Clebsch-Gordan coefficients allow to express the $|j,m\rangle$ vectors for $j=1$ in the Dicke basis $\{|a,b\rangle \equiv |D_5^{(a)}\rangle \otimes |D_5^{(b)}\rangle\,, 0\leqslant a,b\leqslant 5\}$ as

$$|1,1\rangle = \frac{1}{\sqrt{7}}\left(|0,4\rangle+|4,0\rangle\right) - \sqrt{\frac{8}{35}}\left(|1,3\rangle+|3,1\rangle\right) + \frac{3}{\sqrt{35}}|2,2\rangle,$$
$$|1,0\rangle = \sqrt{\frac{5}{14}}\left(|0,5\rangle+|5,0\rangle\right) - \frac{3}{\sqrt{70}}\left(|1,4\rangle+|4,1\rangle\right) + \frac{1}{\sqrt{70}}\left(|2,3\rangle+|3,2\rangle\right),$$
$$|1,-1\rangle = \frac{1}{\sqrt{7}}\left(|1,5\rangle+|5,1\rangle\right) - \sqrt{\frac{8}{35}}\left(|2,4\rangle+|4,2\rangle\right) + \frac{3}{\sqrt{35}}|3,3\rangle. \quad \text{(L13)}$$

Projecting the tensor product $|\Psi\rangle = |\psi\rangle\otimes|\psi\rangle$ on the corresponding subspace gives

$$\langle 1,1|\Psi\rangle = \sqrt{\frac{5}{7}}g, \quad \langle 1,0|\Psi\rangle = -\sqrt{\frac{5}{14}}h, \quad \langle 1,-1|\Psi\rangle = \sqrt{\frac{5}{7}}f, \quad \text{(L14)}$$

thus the components of Y can be interpreted as the projection of the double copy state onto the spin-1 part.

In order to find extrema of S_2 (or $S_2/\binom{N}{2}$) at fixed S_1, we consider the Lagrangian

$$\mathcal{L} = \mathrm{tr}\, Q^2 - \lambda\left(\langle\psi|\psi\rangle - 1\right) - \mu\left(\|v\|^2 - \frac{c}{N}\right) \quad \text{(L15)}$$

with λ,μ two Lagrange multipliers fixing the constraints of normalization and the value $S_1=c$. For $\psi\to\psi+\delta\psi$, the Lagrangian transforms to $\mathcal{L}+\delta\mathcal{L}$, with

$$\delta\mathcal{L} = 2\mathrm{tr}\left(Q\delta Q\right) - \lambda\left(\langle\delta\psi|\psi\rangle + \langle\psi|\delta\psi\rangle\right) - 2\mu\sum_a v_a\delta v_a. \quad \text{(L16)}$$

We can rewrite (L16) as

$$\delta\mathcal{L} = 2\,\langle\delta\psi|\,H\,|\psi\rangle - \lambda\,\langle\delta\psi|\psi\rangle - 2\mu\,\langle\delta\psi|\,W\,|\psi\rangle + \text{c.c.} \quad \text{(L17)}$$

with operators W and H defined in (L7)–(L8).

Since v_a and Q_{ab} are real numbers, H and W are Hermitian matrices. The stationarity condition for the Lagrangian thus reads

$$2(H-\mu W)\,|\psi\rangle = \lambda\,|\psi\rangle\,. \quad \text{(L18)}$$

A stationary point is therefore necessarily an eigenstate of $H-\mu W$.

For odd N, time reversal gives $\langle\tilde{\psi}|\,L_{a_1\ldots a_m 0\ldots 0}\,|\psi\rangle = 0$ for even m. In particular, for $N=5$, $\langle\tilde{\psi}|\psi\rangle = \langle\tilde{\psi}|\,H\,|\psi\rangle = 0$. Equation (L18) therefore implies $\mu\,\langle\tilde{\psi}|\,W\,|\psi\rangle = 0$; inferring $\langle\tilde{\psi}|\,W\,|\psi\rangle = 0$ requires $\mu\neq 0$.

From the stationarity condition (L18), $2H\,|\psi\rangle = (2\mu W+\lambda)\,|\psi\rangle$. Using hermiticity of H and W, X_c can be expressed as

$$\begin{aligned} X_c &= \langle\psi|\,L_c(2\mu W+\lambda)\,|\psi\rangle + \langle\psi|\,(2\mu W+\lambda)L_c\,|\psi\rangle \\ &= 2\mu\textstyle\sum_a v_a\,\langle\psi|\,L_cL_a + L_aL_c\,|\psi\rangle + 2\lambda\,\langle\psi|\,L_c\,|\psi\rangle \end{aligned} \quad \text{(L19)}$$

Using Eqs. (L4) and (L5) we have

$$\begin{aligned} L_cL_a + L_aL_c &= \frac{4}{25}\left(J_aJ_c + J_cJ_a\right) \\ &= \frac{8}{5}\left(L_{ac} + \frac{\delta_{ac}}{4}\mathbf{1}\right) \end{aligned} \quad \text{(L20)}$$

thus (L19) becomes

$$X_c = \frac{16}{5}\mu\sum_a v_aQ_{ac} + \left(2\lambda+\frac{4}{5}\mu\right)v_c \quad \text{(L21)}$$

or equivalently, since Q is a symmetric matrix,

$$X = \frac{16}{5}\mu\,Qv + \left(2\lambda+\frac{4}{5}\mu\right)v. \quad \text{(L22)}$$

We have thus decomposed the 3-dimensional vector X onto vectors v and Qv. This is to be compared with Eq. (L11), which holds for arbitrary $|\psi\rangle$.

We now restrict to the branch $c>0$, $\mu\neq 0$, and rotate so that $v_1=v_2=0$ and $v_3=\sqrt{c/5}>0$. In that case, the condition $\langle\tilde{\psi}|\,W\,|\psi\rangle = 0$ gives $v_3\,\langle\tilde{\psi}|\,L_3\,|\psi\rangle = 0$. Since the constraint $N\|v\|^2 = c$ also needs to be fulfilled, for $c\neq 0$ we must have $v_3\neq 0$. Thus we get the necessary condition $\langle\tilde{\psi}|\,L_3\,|\psi\rangle = 0$, i.e. $Y_3 = -ih = 0$. The vector Y in (L11) thus lies in the (x,y) plane, so that the cross product $Y\times Y^*$ is along $\hat{z}$. Since vector v is also along $\hat{z}$, subtracting (L22) from (L11) gives for normalized states

$$\frac{16}{5}(\mu-1)\,Qv + \left(2\lambda+\frac{4}{5}(\mu-1)\right)v = \frac{4i}{5}Y\times Y^* \quad \text{(L23)}$$

Thus $(\mu-1)Qv$ is also along the z-axis. This leads either to $\mu=1$ or Qv parallel to $\hat{z}$.

If Qv is along $\hat{z}$, then $Q_{31}=Q_{13}=Q_{23}=Q_{32}=0$ (since v is along $\hat{z}$). The 2×2 upper left block of matrix Q can be diagonalized in the (x,y) plane by a rotation Λ with axis $\hat{z}$. This means that in the SU(2) orbit of an extremal state $|\psi\rangle$ there is an extremal state with diagonal matrix $Q_{ab}=q_a\delta_{ab}$ and vector v along $\hat{z}$. Suppose we now consider such an extremal state. Then we have $W=v_3L_3$ and $H=\sum_a q_aL_{aa}$ where, from (L4) and (L5), $L_3=\frac{2}{5}J_z$ and $L_{aa}=\frac{1}{5}J_a^2-\frac{1}{4}\mathbf{1}$. In that case, the 6×6 matrix $2(H-\mu W)$ in Eq. (L18) splits into two blocks, one spanned by the even sector $\{|D_N^{(2i)}\rangle\,, i=0,1,2\}$ and the other by the odd sector $\{|D_N^{(2i+1)}\rangle\,, i=0,1,2\}$. If the stationary eigenvalue of $H-\mu W$ is nondegenerate, its eigenvector lies in one parity sector and has C_2 symmetry. A degeneracy between the two blocks can admit mixtures of the sectors and requires separate analysis.

Within the branch $c>0$, $\mu\neq 0$, the remaining case is $\mu=1$. From (L23) this corresponds to

$$\lambda\,v = \frac{2i}{5}Y\times Y^*, \quad \text{(L24)}$$

with $Y=(i(f-g),f+g,0)$ since $h=0$. Since $v\parallel\hat{z}$, the constraint $N\|v\|^2=c$ gives $v_3^2=c/5$. Thus (L24) gives

$$\frac{2i}{5}(Y\times Y^*)_3 = \frac{4}{5}\left(|g|^2-|f|^2\right) = \lambda\sqrt{\frac{c}{5}} \quad \text{(L25)}$$

at an extremum of S_2 with fixed $S_1=c$. Using Eq. (L14), the necessary conditions on $|\Psi\rangle=|\psi\rangle\otimes|\psi\rangle$ are

$$|\,\langle 1,1|\Psi\rangle\,|^2 - |\,\langle 1,-1|\Psi\rangle\,|^2 = \frac{25}{28}\lambda\sqrt{\frac{c}{5}}, \quad \text{(L26)}$$
$$\langle 1,0|\Psi\rangle = 0. \quad \text{(L27)}$$

These stationarity conditions alone do not prove that every extremizer has C_2 symmetry. The cases $c = 0$, $\mu = 0$, $\mu = 1$, possible degeneracies between the parity blocks, and any points where the constraint gradients are dependent require separate treatment before claiming global optimality of the boundary families.

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